

THE ERDŐS–HAJNAL PROPERTY FOR THE SIX-VERTEX PATH

CLAUDE (ANTHROPIC)

ABSTRACT. We prove that there is $\tau > 0$ such that every graph on n vertices with no induced path on six vertices has a clique or a stable set with at least n^τ vertices; that is, the six-vertex path P_6 has the Erdős–Hajnal property. The proof follows the proof of Nguyen, Scott and Seymour for the five-vertex path and replaces the two steps of that proof which use the absence of an induced house. One replacement is structural: a *module law* forced by the absence of the complement of P_6 , combined with the modular decomposition of the teeth of a comb (the *Tooth Lemma*), shows that combs can be cleaned into blockades whose pairs are complete or sparse on average, and the layout theorem of Nguyen, Scott and Seymour accepts such blockades. The other replacement shows that the blocks on which a vertex is lightly mixed induce a house-free pattern, so that the Erdős–Hajnal theorem for P_5 can be applied to the pattern. The whole argument has been formalized in Lean 4 with Mathlib. The formalization also proves one of the four cited theorems, takes the other three as hypotheses, and has been checked both by Lean’s kernel and by an independent type checker.

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INTRODUCTION

A graph G is H -free if no induced subgraph of G is isomorphic to H . Write $\text{hom}(G)$ for the largest size of a clique or a stable set of G . Every graph on n vertices has $\text{hom}(G) \geq \frac{1}{2} \log_2 n$, and

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This paper, the proof it contains and the accompanying Lean formalization were produced by an AI system in a research session commissioned by the owner of the repository <https://github.com/machine-qed/erdos-hajnal-six-vertex-path>, who is responsible for their release. It is not a publication of Anthropic, and Anthropic has not reviewed it. The paper has not been refereed. Please report problems as issues in that repository. The paper is released under CC BY 4.0.

random graphs show that this is best possible up to the constant. In 1977 Erdős and Hajnal [5, 6] conjectured that excluding a single induced subgraph changes the picture completely.

Conjecture (Erdős–Hajnal). *For every graph H there exists $c > 0$ such that every H -free graph G satisfies $\text{hom}(G) \geq |G|^c$.*

A graph H for which this holds is said to have the *Erdős–Hajnal property*. The property is known for every graph with at most four vertices and is preserved by substitution [1]. The bull was settled by Chudnovsky and Safra [3], and the five-cycle by Chudnovsky, Scott, Seymour and Spirkl [4]. Nguyen, Scott and Seymour proved it for the five-vertex path [10], which settled every graph on five vertices. Since then it has been proved for a few six-vertex graphs: the E-graph and the bird [8], and the graph with edges $ab, bc, cd, de, af, bf, df$ [12]. For paths in general, Nguyen, Scott and Seymour [9] showed that P_k -free graphs satisfy $\text{hom}(G) \geq 2^{(\log |G|)^{1-o(1)}}$, and Bousquet, Lagoutte and Thomassé [2] proved a polynomial bound when both P_k and its complement are excluded. After P_5 , the six-vertex path P_6 is the smallest path for which the conjecture was open.

Theorem A. *There exists $\tau > 0$ such that every P_6 -free graph G satisfies $\text{hom}(G) \geq |G|^\tau$.*

A graph is P_6 -free if and only if its complement is $\overline{P_6}$ -free, and $\text{hom}(G) = \text{hom}(\overline{G})$, so Theorem A is equivalent to the Erdős–Hajnal property of $\overline{P_6}$. We work with $\overline{P_6}$ -free graphs throughout. We in fact prove the stronger *polynomial Rödl property* (Theorem 8.3): there is A such that for every $\varepsilon \in (0, \frac{1}{2})$, every $\overline{P_6}$ -free graph G has an induced subgraph on at least $\varepsilon^{3A}|G|$ vertices whose maximum degree, or whose complement’s maximum degree, is at most ε times its number of vertices.

The proof determines τ by a formula (Table 1) in the constant δ of Rödl’s theorem and the exponent of the Erdős–Hajnal theorem for P_5 . No numerical values of these are available, so we cannot give τ numerically; the formula only shows that $\tau < 10^{-15}$.

What is new and what is inherited. The proof uses four results from the literature as black boxes: Rödl’s theorem, a blockade theorem of Nguyen, Scott and Seymour for sparse graphs without a long induced path, the Erdős–Hajnal theorem for P_5 , and the comb lemma of Chudnovsky, Scott, Seymour and Spirkl. The architecture of the proof (iterative sparsification organised around combs, niceness and a layout theorem) is that of the proof for P_5 by Nguyen, Scott and Seymour [10]. That proof uses the absence of an induced *house*, the complement of P_5 , in exactly two places, and both fail for $\overline{P_6}$ -free graphs. The following parts are new:

- the three six-vertex certificates: the module law (Lemma 1.1) and the diagonal and house lemmas (Lemmas 1.2 and 1.3);
- the Tooth Lemma (Lemma 2.1), which combines the module law with the modular decomposition of a comb’s teeth;
- the resulting comb lemma for $\overline{P_6}$ (Lemma 3.1), whose output blockades are only *semisparse*, and the observation that the layout theorem accepts semisparse input (Theorem 5.1);
- the house-free-pattern argument that brings the Erdős–Hajnal theorem for P_5 into the second sparsification round (Corollary 1.4 and Lemma 6.1).

The remaining steps, namely the two sparsification rounds and the reduction to the polynomial Rödl property, follow [10] with the constants recomputed. We prove them in full as well, in the form in which they were formalized.

The imported theorems. The proof uses the following four theorems and nothing else. Section 0.2 quotes each from its source, next to the Lean proposition that renders it. Theorems I, III and IV come from refereed papers. Theorem II comes from [9], which has not yet appeared

in a journal; its proof takes half a page, and the formalization proves it, so the formal result assumes only Theorems I, III and IV.

	imported theorem	Lean proposition
I	Rödl’s theorem [11], as stated in [10, Theorem 1.3], for $H = \overline{P_6}$	Rod1CoP6
II	sparse graphs without an induced path P contain long anticomplete blockades [9, statement 3.1], for $P = P_6$	NssPath6
III	the Erdős–Hajnal property of P_5 [10, Theorem 1.2]	EhP5
IV	the comb lemma [10, Lemma 4.3], from [4]	NssComb

Formal verification. The whole argument is formalized in Lean 4 [19] with Mathlib [18]. The formal theorem `EHP6.erdos_hajnal_P6_of_imported` states that the four propositions above imply Theorem A. Theorem II is proved in the formalization, so Theorems I, III and IV alone imply Theorem A; this is the formal theorem `erdos_hajnal_P6_of_cited`. The proofs of both use no axioms beyond Lean’s three standard ones. It was checked with Lean 4.34.1, replayed from an empty environment with `leanchecker`, and checked by the comparator tool with two kernels, Lean’s own and nanoda, an independent type checker written in Rust. The source and build instructions are in the repository <https://github.com/machine-qed/erdos-hajnal-six-vertex-path>. Section 9 describes the definitions, the correspondence between the paper and the Lean files, and how to reproduce the checks. The paper can be read without any knowledge of Lean. Independently of this work, Bonnet has formalized in Lean the Erdős–Hajnal theorems for C_5 and P_5 , including Rödl’s theorem and a version of the comb lemma [16, 17]. Connecting these to our formalization would turn Theorems I and III, and, after a change of constant, IV into proved statements; we have not done this.

Priority. To the best of our knowledge, Theorem A is new. On 27 September 2026 we checked every paper indexed by Semantic Scholar that cites [9], [10], [4] or [8], searched arXiv and the web, and read the page and discussion of the conjecture on the Erdős problems website (Problem 61), Paul Seymour’s list of papers, and the lists of Erdős problems resolved with the help of AI. We found no proof of the Erdős–Hajnal property for P_6 . The closest recent results are [8] and [12]. By the count of Tran and Nguyen [12], eight complementary pairs of prime six-vertex graphs were open as of 26 August 2026, and $\{P_6, \overline{P_6}\}$ is one of them. A public repository by Jared Wilder [15] develops local structure for P_6 -free graphs towards the same goal and states that it does not prove the full result. We did not have access to MathSciNet, zbMATH or Google Scholar, and we cannot rule out unpublished work.

Organization. The next section explains the idea of the proof and where $\overline{P_6}$ -freeness is used. Section 0 fixes notation, states the four imported theorems, and proves the elementary tools. Sections 1–3 contain the new structural part: the local lemmas, the Tooth Lemma and the comb lemma. Sections 4–6 run the two sparsification rounds. Sections 7 and 8 derive Theorem A, and Section 9 documents the formalization.

THE IDEA OF THE PROOF

Iterative sparsification. Following Nguyen, Scott and Seymour, we reduce the Erdős–Hajnal property to a statement about sparse graphs, Crux (C) (Definition 7.1): every y -sparse $\overline{P_6}$ -free graph contains either a large induced subgraph that is much sparser, namely y^2 -sparse, or a long complete or anticomplete blockade. Passing repeatedly to a much sparser large subgraph either stops at a long complete or anticomplete blockade or produces a very sparse set of polynomial size, and an argument of Nguyen, Scott and Seymour (Section 8) turns this into the polynomial Rödl property and then into Theorem A. Crux (C) is proved in two *sparsification rounds*.

Round one (Sections 3–5) proves that P_6 is *nice*: every $\overline{P_6}$ -free graph contains a long blockade of polynomial width whose pairs are complete or sparse. Round two (Section 6) turns niceness into Crux (C).

Where the house was used, and what replaces it. The proof for P_5 uses house-freeness twice, and both uses fail for $\overline{P_6}$ -free graphs.

Round one: combs. Around a vertex v of large degree one finds a *comb*: vertices $a_1, \dots, a_\ell \notin N(v)$ and disjoint *teeth* $B_i \subseteq N(v)$ such that a_i is complete to B_i and anticomplete to every other tooth (Figure 3). For house-free graphs, a vertex of one tooth cannot be mixed on another tooth, so the teeth are pure to each other. In a $\overline{P_6}$ -free graph this is false. What survives is the *module law* (Lemma 1.1): for a vertex u of another tooth, each anticomponent of $N(u) \cap B_i$ is a *module* of B_i . Otherwise six vertices form an induced $\overline{P_6}$ (Figure 1(a)). So an outside vertex sees a tooth only through the tooth’s modular structure. The *Tooth Lemma* (Lemma 2.1) exploits this through the modular decomposition tree of the tooth. Each tooth contains a long pure or complete blockade, or a large subset Y' that every other tooth’s vertices see either completely or very sparsely. In the second case the Pair Lemma shows that any two such subsets are complete or *weakly sparse*, that is, sparse on average. The resulting blockades are *semisparse* rather than pure. The last observation of round one is that the layout theorem of Nguyen, Scott and Seymour (Theorem 5.1) only counts “wrong” pairs, and a weakly sparse pair contributes few of them. So semisparse input suffices, and P_6 is nice.

Round two: mixed vertices. Given a long blockade of anticonnected blocks whose pairs are complete or sparse, the proof for P_5 uses the house to show that no outside vertex is mixed on two complete blocks. For $\overline{P_6}$ -free graphs a vertex can be mixed on many blocks, and the pattern of those blocks can be complicated. However, the pattern cannot contain a house. A house in the pattern of the blocks on which v is lightly mixed yields an induced $\overline{P_6}$ through v (the house lemma, Lemma 1.3; Figure 1(c)). One configuration inside the house must be shown anticomplete first, which is the job of the diagonal lemma (Lemma 1.2; Figure 1(b)). A house-free graph is the complement of a P_5 -free graph, so the Erdős–Hajnal theorem for P_5 applies to the pattern and gives polynomially many blocks that are pairwise complete or pairwise sparse. The first is a long complete blockade; the second is a much sparser set. Either way the round goes through.

In short, the new part of the argument is the chain

$$\begin{aligned} \text{module law} &\rightarrow \text{Tooth Lemma} \rightarrow \text{semisparse comb} \rightarrow \text{semisparse layout} \\ &\rightarrow \text{house-free block pattern} \rightarrow \text{EH}(P_5). \end{aligned}$$

Where the six-vertex path enters. In each of the three places, the structure built so far produces six vertices with prescribed adjacencies: ten pairs are adjacent because of completeness between blocks or membership in a neighbourhood, and five are non-adjacent because of how the vertices were chosen. In all three configurations the five non-adjacent pairs form a path (Figure 1), so the six vertices induce $\overline{P_6}$, which the hypothesis rules out. The hypothesis cannot simply be weakened: in the graph $\overline{P_6}$ itself the module law fails, with $Z = \{y, z, w\}$ as in Figure 1(a). Since $\overline{P_6}$ has no induced $\overline{P_7}$, the method does not extend to P_7 without new ideas.

0. PRELIMINARIES

0.1. Notation. Graphs are finite and simple, $|G| = |V(G)|$, $N(v)$ is the neighbourhood of v , and $G[X]$ is the subgraph induced on $X \subseteq V(G)$. We freely identify X with $G[X]$. For disjoint A, B , $e(A, B)$ is the number of edges between them. A is *complete* (*anticomplete*) to B if every (no) pair ab with $a \in A$, $b \in B$ is an edge. The pair is *pure* if it is one of the two. A vertex is *mixed* on a set if it has both a neighbour and a non-neighbour there.

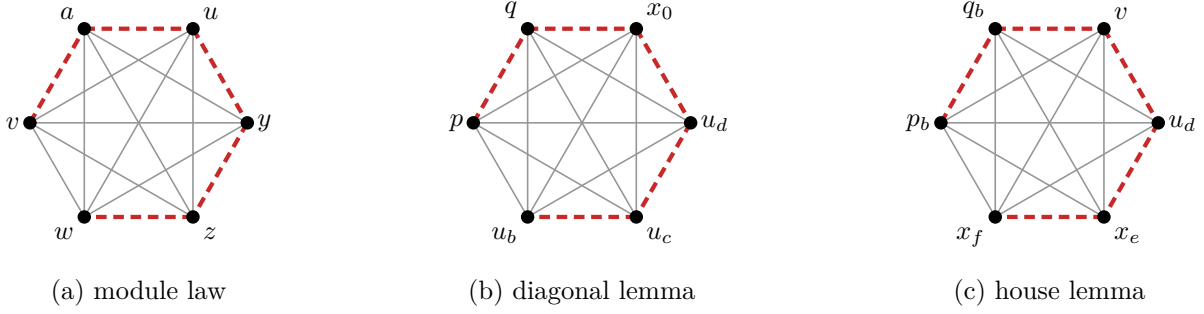


FIGURE 1. The three six-vertex certificates. Dashed red: the five non-adjacent pairs, which form a path. Grey: the ten adjacent pairs. Each configuration is therefore an induced $\overline{P}_6$. In (a), v, a are the root and a handle of a comb, u lies in another tooth, y is a tooth vertex outside $N(u)$ mixed on an anticomponent K of $N(u) \cap B_i$, and $w \in N(y) \cap K$, $z \in K \setminus N(y)$ are non-adjacent. In (b) and (c) the subscripts name the blocks $B_b, \dots, B_f$ of a house in the pattern graph. In (c), v is the lightly mixed outside vertex.

Definition 0.1 (Sparsity). Let $x \geq 0$. A graph G is x -sparse if every vertex has at most $x|G|$ neighbours, and x -restricted if G or $\overline{G}$ is x -sparse. A set X is x -sparse or x -restricted if $G[X]$ is. For disjoint A, B : B is x -sparse to A if every vertex of B has at most $x|A|$ neighbours in A , and (A, B) is weakly x -sparse if $e(A, B) \leq x|A||B|$. If B is x -sparse to A then (A, B) is weakly x -sparse.

Definition 0.2 (Blockades). For reals k, w , a (k, w) -blockade is a sequence $(B_1, \dots, B_m)$ of pairwise disjoint sets (blocks, possibly empty) with $m \geq k$ and $|B_i| \geq w$ for all i . It is complete, anticomplete or pure if every two blocks are complete, anticomplete or pure, respectively. It is x -sparse if B_j is x -sparse to B_i whenever $i < j$, and x -semisparse if every two blocks are complete or weakly x -sparse. A pure blockade is 0-semisparse; an x -sparse blockade is x -semisparse.

A set X is *anticonnected* if $\overline{G}[X]$ is connected, equivalently if every partition of X into two nonempty parts has a non-adjacent pair across. An *anticomponent* of X is the vertex set of a component of $\overline{G}[X]$. Distinct anticomponents of X are complete to each other, and each anticomponent is complete to the rest of X . We write $\text{hom}(G) = \max(\alpha(G), \omega(G)) = \text{hom}(\overline{G})$. P_k is the path on k vertices, $\overline{P}_6$ is its complement for $k = 6$, namely six vertices whose five non-adjacent pairs form a path, and the *house* is $\overline{P}_5$. Logarithms are to base 2.

0.2. The four imported theorems. We quote each result as stated in its source, together with the definitions it depends on, and then give the Lean proposition that renders it (file `formal/EHP6/Imported.lean`). The main formal theorem takes these four propositions as hypotheses. Theorem II is proved in the formalization, so the unconditional formal result assumes only the other three; see Section 9. Each proposition is stated for an arbitrary vertex set S of a graph G , meaning the induced subgraph $G[S]$. This is equivalent to the source statement because induced subgraphs of H -free graphs are H -free. In the Lean code, `Free G H` means that G has no induced copy of H , `Sparse G x S` means that every vertex of S has at most $x|S|$ neighbours in S , and `Restricted G x S` means `Sparse G x S` or `Sparse Gc x S`. Section 9 gives the full definitions.

Imported Theorem I (Rödl [11]; stated as [10, Theorem 1.3]). Source: “For every $\varepsilon \in (0, 1/2)$ and every graph H , there exists $\delta > 0$ such that every H -free graph G has an ε -restricted induced subgraph with at least $\delta|G|$ vertices.” Here G is ε -sparse if its maximum degree is at most $\varepsilon|G|$,

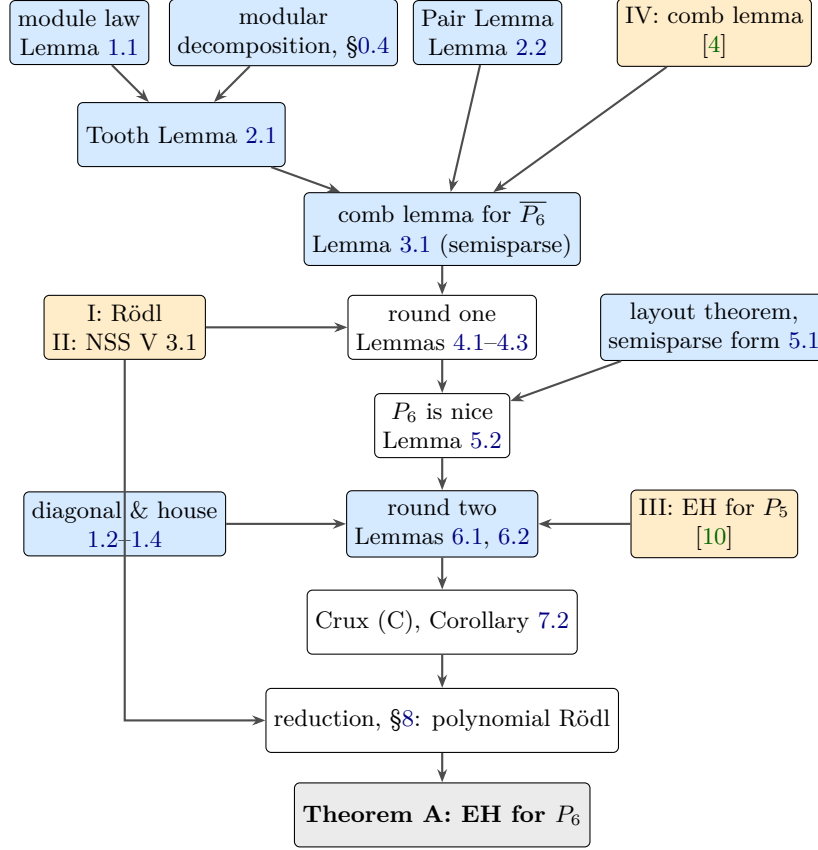


FIGURE 2. Structure of the proof. Blue: proved here and specific to $\overline{P_6}$ or new. Orange: the four imported theorems (II is proved in the formalization). White: steps that follow [10], with constants recomputed.

and ε -restricted if one of $G, \overline{G}$ is ε -sparse. We use it only for $H = \overline{P_6}$ and write $\delta(\varepsilon)$ for the constant.

```

def RodlCoP6 : Prop := ∀ ε : ℝ, 0 < ε → ε < 1 / 2 → ∃ δ : ℝ, 0 < δ ∧
  ∀ (V : Type) [Fintype V] [DecidableEq V] (G : SimpleGraph V) [DecidableRel G.Adj],
    Free G P6 → ∀ S : Finset V, ∃ F ⊆ S, δ * S.card ≤ F.card ∧ Restricted G ε F

```

Imported Theorem II ([9, statement 3.1]). Source: “Let P be a path with $k \geq 2$ vertices, and let $0 < y \leq 1/(60k)$. Let G be a y^2 -sparse graph. Then either G contains a copy of P , or G contains an anticomplete $(1/y, \lfloor y^2|G| \rfloor)$ -blockade in G .” In [9] a blockade is a finite sequence of possibly empty disjoint sets, and a (k, w) -blockade has length at least k and width at least w , as above. A “copy” of P is an induced copy: H -freeness in [9] refers to induced subgraphs, and the proof of [9, 3.1] constructs an induced path. We use it for $P = P_6$, so $k = 6$ and $y \leq 1/360$.

```

def NssPath6 : Prop := ∀ y : ℝ, 0 < y → y ≤ 1 / 360 →
  ∀ (V : Type) [Fintype V] [DecidableEq V] (G : SimpleGraph V) [DecidableRel G.Adj]
    (S : Finset V), Sparse G (y ^ 2) S → ¬ ContainsInduced G P6 S →
    ∃ β : Blockade S (1 / y) (⌊ y ^ 2 * S.card ⌋ + : ℝ), β.IsAnticomplete G

```

Proved in the formalization (*formal/EHP6/NSSPath.lean*, for every $k \geq 1$), following the proof in [9]. The only change is that the repeated removal of small sets with few neighbours is replaced

by a maximal family of such sets, which are pairwise disjoint and anticomplete; the maximality gives the expansion property of the remaining vertices directly.

Imported Theorem III (Nguyen, Scott and Seymour [10, Theorem 1.2]). Source: “ P_5 satisfies the Erdős–Hajnal conjecture,” that is, there exists $c > 0$ such that every P_5 -free graph G has a clique or a stable set of size at least $|G|^c$. We fix such a $\kappa > 0$. Since $\text{hom}(G) = \text{hom}(\overline{G})$, every house-free graph G also has $\text{hom}(G) \geq |G|^\kappa$.

```
def EhP5 : Prop := ∃ c : ℝ, 0 < c ∧
  ∀ (V : Type) [Fintype V] [DecidableEq V] (G : SimpleGraph V) [DecidableRel G.Adj],
    Free G P5 → ∃ T : Finset V, (IsCliqueF G T ∨ IsStableF G T) ∧ (Fintype.card V : ℝ) ^ c
    ≤ T.card
```

Imported Theorem IV (the comb lemma; [10, Lemma 4.3], a special case of a lemma of [4]). Source: “Let G be a graph and let $A, B \subseteq V(G)$ be nonempty and disjoint, such that each vertex in A has at most $\Delta > 0$ neighbours in B . Then either at most $20\sqrt{|B|\Delta}$ vertices in B have a neighbour in A ; or for some integer $k \geq 1$, there is a $(k, |B|/k^2)$ -comb $((a_i, B_i) : i \in [k])$ in G where $a_i \in A$ and $B_i \subset B$ for all $i \in [k]$.” Here an (ℓ, w) -comb is a sequence of pairs $((a_i, B_i) : i \in [\ell])$ where $(B_1, \dots, B_\ell)$ is an (ℓ, w) -blockade, the a_i are pairwise distinct, $\{a_1, \dots, a_\ell\}, B_1, \dots, B_\ell$ are pairwise disjoint, and for distinct i, j , a_i is complete to B_i and anticomplete to B_j . The symbol $\subset$ in the source denotes inclusion; for $k = 1$ it must allow $B_1 = B$.

```
def NssComb : Prop :=
  ∀ (V : Type) [Fintype V] [DecidableEq V] (G : SimpleGraph V) [DecidableRel G.Adj]
    (A B : Finset V) (Δ : ℝ), Disjoint A B → A.Nonempty → B.Nonempty → 0 < Δ →
    (∀ a ∈ A, ((nbrs G a B).card : ℝ) ≤ Δ) →
    (((B.filter (fun b => ∃ a ∈ A, G.Adj a b)).card : ℝ) ≤ 20 * Real.sqrt (B.card * Δ)) ∨
    (∃ ℓ : ℕ, 1 ≤ ℓ ∧ ∃ (a : Fin ℓ → V) (C : Fin ℓ → Finset V),
      Function.Injective a ∧ (∀ i, a i ∈ A) ∧ (∀ i, C i ⊆ B) ∧
      (∀ i j, i ≠ j → Disjoint (C i) (C j)) ∧
      (∀ i, (B.card : ℝ) / ℓ ^ 2 ≤ (C i).card) ∧
      (∀ i, ∀ c ∈ C i, G.Adj (a i) c) ∧
      (∀ i j, i ≠ j → ∀ c ∈ C j, ¬ G.Adj (a i) c))
```

The disjointness of the apexes from the teeth follows from $A \cap B = \emptyset$.

0.3. Two elementary lemmas of Nguyen, Scott and Seymour. The proof also uses Lemmas 4.1 and 4.2 of [10]. We prove both, since they are short and are proved in the formalization. Lemma 0.4 is [10, Lemma 4.2] in the form of arXiv version 3 (February 2026), with the added hypothesis $A \neq \emptyset$: for $A = \emptyset \neq B$ the hypothesis holds vacuously but the conclusion fails. This does not affect any use of it, here or in [10]. The revision of [10] dated July 2026 states the lemma for all $x > 0$ with the weaker bound $|A'| \leq 2/x$; since we prove Lemma 0.4 below, nothing here depends on which version is used.

Lemma 0.3 ([10, Lemma 4.1]). Let $k \geq 2$ be an integer and let G be a graph all of whose anticomponents have fewer than $|G|/k$ vertices. Then G has a complete $(k, |G|/k^2)$ -blockade.

Proof. Let $n = |G|$, $w = n/k^2$, $c = n/k$. Distinct anticomponents are complete to each other, so if $\mathcal{G}_1, \dots, \mathcal{G}_k$ are disjoint collections of anticomponents, the unions $\bigcup \mathcal{G}_1, \dots, \bigcup \mathcal{G}_k$ are pairwise complete. From any collection of anticomponents of total size at least w we can pick a subcollection of total size in $[w, w + c)$: add members one at a time until the total reaches w , and use that every member has size $< c$. Pick such subcollections $\mathcal{G}_1, \mathcal{G}_2, \dots$ greedily from the anticomponents not yet used. After $j \leq k - 1$ steps at most $(k - 1)(w + c) = n - n/k^2$ vertices are used, so the remaining anticomponents have total size at least w and step $j + 1$ is possible. The unions of $\mathcal{G}_1, \dots, \mathcal{G}_k$ form a complete $(k, n/k^2)$ -blockade. $\square$

Applied to $\overline{G}$: if every component of G has fewer than $|G|/k$ vertices, then G has an anticomplete $(k, |G|/k^2)$ -blockade.

Lemma 0.4 (cf. [10, Lemma 4.2]). *Let $x \in (0, \frac{1}{2})$, let $A \neq \emptyset$ and B be sets of vertices, and suppose every vertex of B has at least $x|A|$ neighbours in A . Then there is $A' \subseteq A$ with $|A'| \leq 1/x$ such that at least $|B|/2$ vertices of B have a neighbour in A' .*

Proof. Let $U \subseteq B$. Counting edges, $\sum_{a \in A} |N(a) \cap U| = \sum_{b \in U} |N(b) \cap A| \geq x|A||U|$, so some $a \in A$ has at least $x|U|$ neighbours in U . Choose $a_1, a_2, \dots, a_t$ greedily, each covering an x -fraction of the vertices of B not yet covered, where $t = \lfloor 1/x \rfloor$. At most $(1-x)^t|B|$ vertices remain uncovered. As $x < \frac{1}{2}$, $t \geq 2$ and $x > 1/(t+1)$, so $1-x < t/(t+1)$, and Bernoulli's inequality $(1+1/t)^t \geq 2$ gives $(1-x)^t \leq (t/(t+1))^t \leq \frac{1}{2}$. Take $A' = \{a_1, \dots, a_t\}$. $\square$

Lemma 0.5. *If D is anticonnected and $1 \leq b \leq |D|$, then D contains an anticonnected set of exactly b vertices.*

Proof. Start from a single vertex. If $B \subsetneq D$ is anticonnected, anticonnectivity of D applied to the partition $(B, D \setminus B)$ gives $p \in B$ and $q \in D \setminus B$ with $pq \notin E(G)$, and $B \cup \{q\}$ is anticonnected. $\square$

0.4. Modular decomposition. Let Y be a set of vertices. A *module* of $G[Y]$ is a set $M \subseteq Y$ such that every vertex of $Y \setminus M$ is complete or anticomplete to M . A module is *strong* if it overlaps no other module, where two sets overlap if they meet and neither contains the other. The following facts are standard (Gallai [7]); the forms used below are proved in the formalization. The strong modules of $G[Y]$ form a laminar family containing Y and all singletons, that is, a rooted tree under inclusion. The *children* of a strong module N with $|N| \geq 2$ are the maximal strong modules properly contained in N , and they partition N . There are three kinds of node:

- N is *parallel* if $G[N]$ is disconnected; then its children are the components of $G[N]$;
- N is *series* if $\overline{G}[N]$ is disconnected; then its children are the anticomponents of N ;
- N is *prime* if $G[N]$ and $\overline{G}[N]$ are both connected; then the quotient graph on the children (two children adjacent if complete to each other) is prime, that is, it has at least four vertices and no modules other than the trivial ones.

A module of $G[N]$ for a module N of $G[Y]$ is a module of $G[Y]$.

Lemma 0.6 (MD1). *Disjoint modules A, B of $G[Y]$ are pure to each other.*

Proof. Each $a \in A$ lies outside B , so it is complete or anticomplete to B . If $a_1 \in A$ were complete and $a_2 \in A$ anticomplete to B , any $b \in B$ would be mixed on the module A , impossible as $b \notin A$. $\square$

Lemma 0.7 (MD2). *If N is a prime node, every module X of $G[N]$ equals N or is contained in one child of N .*

Proof. Let I be the set of children meeting X , and suppose $|I| \geq 2$. For a child $C \notin I$ and $c \in C$, the adjacency of c to a child $M \in I$ equals its adjacency to $X \cap M$ (M is a module and $c \notin M$), which equals its adjacency to X (X is a module and $c \notin X$). Hence I is a module of the prime quotient, and since $|I| \geq 2$ it contains all children. If some child $M_1 \not\subseteq X$, take $m \in M_1 \setminus X$. For every other child M the adjacency of m to M equals its adjacency to $X \cap M$, which equals its adjacency to X . So in the quotient, M_1 is complete or anticomplete to all other children, and these, at least three of them, form a nontrivial module of the prime quotient, a contradiction. Hence every child lies in X , and $X = N$. $\square$

1. THE LOCAL LEMMAS

These three lemmas are the only places where $\overline{P}_6$ -freeness is used, apart from the imported Theorems I and II. Each is proved by exhibiting an induced $\overline{P}_6$ (Figure 1).

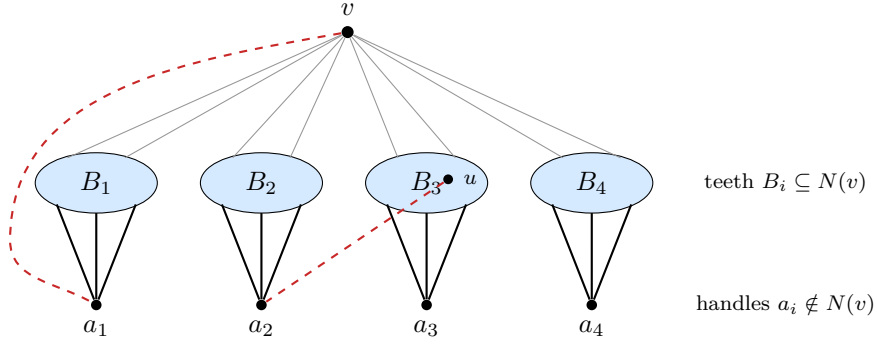


FIGURE 3. A comb with root v . Each handle a_i is complete to its own tooth B_i and anticomplete to the other teeth, and v is complete to every tooth and non-adjacent to the handles. The module law applies to $Z = B_i$ and a vertex u of another tooth B_j : $u \sim v$ and $u \not\sim a_i$.

1.1. The module law.

Lemma 1.1 (module law). *Let G be $\overline{P_6}$ -free, let v, a be non-adjacent vertices, let $Z \subseteq N(v) \cap N(a)$, and let $u \notin Z$ with $u \sim v$ and $u \not\sim a$. Then every anticomponent K of $N(u) \cap Z$ is a module of $G[Z]$.*

Proof. Vertices of $(N(u) \cap Z) \setminus K$ lie in other anticomponents of $N(u) \cap Z$, so they are complete to K . Suppose $y \in Z \setminus N(u)$ is mixed on K . Since K is anticonnected and $N(y) \cap K$ and $K \setminus N(y)$ are both nonempty, there are $w \in N(y) \cap K$ and $z \in K \setminus N(y)$ with $w \not\sim z$. The six vertices v, a, u, y, z, w are distinct: $v, a \notin Z$ because $Z \subseteq N(v) \cap N(a)$, and $u \notin Z$. Their adjacent pairs are v with u, y, z, w ($u \sim v$ by hypothesis, and $y, z, w \in Z$); a with y, z, w ; u with z, w (as $z, w \in K \subseteq N(u)$); and yw . The non-adjacent pairs are va, au, uy, yz, zw , a path. So v, a, u, y, z, w induce $\overline{P_6}$, a contradiction. $\square$

1.2. Mixed-vertex patterns are house-free. Throughout this subsection $B_1, \dots, B_\ell$ is a *block system*: disjoint anticonnected sets of the same size $W \geq 1$ such that, for some $x \in (0, \frac{1}{2})$, every two of them are complete or x -sparse to each other in both directions. Since $x < 1$, no pair is both. The *pattern graph* J on $[\ell]$ has $ij \in E(J)$ if and only if B_i is complete to B_j .

Lemma 1.2 (diagonal lemma). *Let G be $\overline{P_6}$ -free, and let b, c, d, e, f be indices such that B_b is complete to B_d, B_e, B_f , B_c is complete to B_e, B_f , B_d is complete to B_f , and the pairs $(b, c), (c, d), (d, e), (e, f)$ are not complete (hence x -sparse). Then B_e is anticomplete to B_f .*

Proof. Suppose $x_0 \in B_e$ has a neighbour in B_f . Since $|N(x_0) \cap B_f| \leq xW < W$, x_0 also has a non-neighbour in B_f , and since B_f is anticonnected there are $p \in N(x_0) \cap B_f$ and $q \in B_f \setminus N(x_0)$ with $p \not\sim q$. Take $u_b \in B_b$, then $u_c \in B_c \setminus N(u_b)$ (possible as $|N(u_b) \cap B_c| \leq xW < W$), then $u_d \in B_d \setminus (N(x_0) \cup N(u_c))$ (possible as each of x_0, u_c has at most xW neighbours in B_d and $2xW < W$). Among p, q, x_0, u_d, u_c, u_b the adjacent pairs are px_0 (by choice); p and q with u_d, u_c, u_b (as B_f is complete to B_d, B_c, B_b); x_0 with u_c, u_b (as B_e is complete to B_c, B_b); and $u_d u_b$ (as B_d is complete to B_b). The non-adjacent pairs are $pq, qx_0, x_0 u_d, u_d u_c, u_c u_b$, a path. This is an induced $\overline{P_6}$, a contradiction. $\square$

Lemma 1.3 (house lemma). *Let G be $\overline{P_6}$ -free and let b, c, d, e, f be as in Lemma 1.2. Then no vertex $v \notin \bigcup_i B_i$ has all of the following: a neighbour and a non-neighbour in B_b , a neighbour in B_e , a neighbour in B_f , and more than xW non-neighbours in B_d .*

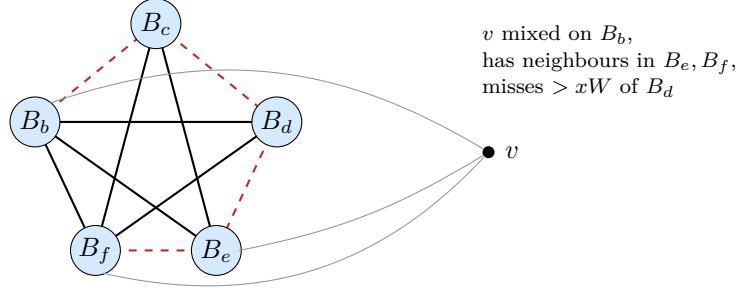


FIGURE 4. A house in the pattern graph: solid pairs are complete, dashed pairs are not complete. The four dashed pairs bc, cd, de, ef form a path, so the pattern is the complement of P_5 . By the house lemma no outside vertex v can be lightly mixed on all five blocks.

Proof. Suppose v has them. Take $x_e \in N(v) \cap B_e$ and $x_f \in N(v) \cap B_f$; by Lemma 1.2, $x_e \not\sim x_f$. By anticonnectivity of B_b there are $p_b \in N(v) \cap B_b$ and $q_b \in B_b \setminus N(v)$ with $p_b \not\sim q_b$. Take $u_d \in B_d \setminus (N(v) \cup N(x_e))$, possible because $|N(x_e) \cap B_d| \leq xW < |B_d \setminus N(v)|$. Among $p_b, q_b, v, u_d, x_e, x_f$ the adjacent pairs are p_b and q_b with u_d, x_e, x_f (as B_b is complete to B_d, B_e, B_f); v with p_b, x_e, x_f ; and $u_d x_f$ (as B_d is complete to B_f). The non-adjacent pairs are $p_b q_b, q_b v, v u_d, u_d x_e, x_e x_f$, a path. This is an induced $\overline{P_6}$, a contradiction. $\square$

For a vertex $v \notin \bigcup_i B_i$ let $I(v) = \{i : 0 < |N(v) \cap B_i| < W/2\}$ be the set of blocks on which v is *lightly mixed*, and $J_v = J[I(v)]$.

Corollary 1.4. *Let G be $\overline{P_6}$ -free and $v \notin \bigcup_i B_i$. Then J_v is house-free. Consequently there is $R \subseteq I(v)$ with $|R| \geq |I(v)|^\kappa$ whose blocks are pairwise complete or pairwise not complete.*

Proof. An induced house in J_v on b, c, d, e, f , with non-edges bc, cd, de, ef (the complement of the path $b-c-d-e-f$), has exactly the completeness pattern of Lemma 1.2. The vertex v is mixed on B_b and has neighbours in B_e and B_f , and $|B_d \setminus N(v)| > W/2 > xW$. This contradicts Lemma 1.3. For the second statement note that the complement of J_v is P_5 -free, so by Theorem III it has a clique or a stable set with at least $|I(v)|^\kappa$ vertices, and so does J_v . A clique of J_v is a set of pairwise complete blocks, and a stable set is a set of pairwise non-complete blocks. $\square$

2. THE TOOTH LEMMA AND THE PAIR LEMMA

The Tooth Lemma is the main new structural result. Its input is a set Y , which will be a tooth of a comb, together with a set U of outside vertices that see Y only through modules, as the module law guarantees. Its output is a long pure or complete blockade inside Y , or a large subset $Y' \subseteq Y$ that each $u \in U$ sees completely or very sparsely.

Lemma 2.1 (Tooth Lemma). *Let $0 < x \leq 2^{-20}$, let k be an integer with $2^{16} \leq k \leq 2x^{-1/2}$, let Y be a set of $n := |Y| \geq 16x^{-3}$ vertices, and let U be a set of vertices such that for every $u \in U$ every anticomponent of $N(u) \cap Y$ is a module of $G[Y]$. Then $G[Y]$ contains one of the following:*

- (TL1) a pure $(K, n/K^4)$ -blockade, for some integer K with $K^4 \geq k$ and $K \leq 1/x$;
- (TL2) a pure $(L, n/(2k))$ -blockade, for an integer L with $(8L)^4 \leq k^3 \leq (9L)^4$;
- (TL3) a complete or anticomplete $(\lfloor 1/x \rfloor, x^2 n/k)$ -blockade;
- (TL4) a complete $(\lfloor 1/(xk) \rfloor, x^3 n/4)$ -blockade;
- (TL5) a set $Y' \subseteq Y$ with $|Y'| \geq n/k$ such that for every $u \in U$, either $Y' \subseteq N(u)$ or $|N(u) \cap Y'| < x|Y'|/4$.

In (TL1) the length is at least $k^{1/4}$, and in (TL2) it lies between $k^{3/4}/9$ and $k^{3/4}/8$. Lengths are written without roots because this is how they are stated in Lean.

Proof. Note that $k \leq 2x^{-1/2}$ gives $k^2x \leq 4$ and $xk \leq 2x^{1/2} \leq 2^{-9}$. Set

$$s := x^2n/4, \quad t := n/k, \quad K_1 := \lfloor 1/x \rfloor.$$

Then $s \geq 4/x > 1$, and $s \leq t$ because $xk \leq 4$. Let $\mathcal{L}$ be the set of strong modules of $G[Y]$ with at least s vertices. It contains Y and is closed upwards in the tree of strong modules. An *atom* is a child of a member of $\mathcal{L}$ that is not itself in $\mathcal{L}$, so every atom has fewer than s vertices. For $N \in \mathcal{L}$ let $S(N)$ be the union of the atom children of N , the *layer* of N . We use three facts.

- (F1) *The layers $S(N)$, $N \in \mathcal{L}$, partition Y .* For $v \in Y$ let N_v be a minimal member of $\mathcal{L}$ containing v . The child of N_v containing v is not in $\mathcal{L}$, so $v \in S(N_v)$. Suppose $v \in S(N) \cap S(N')$ with $N \neq N'$. By laminarity one of them, say N' , is strictly inside the other. The child C of N containing v and the strong module N' both contain v , so they are nested. $C \subsetneq N' \subsetneq N$ would contradict C being a child, so $N' \subseteq C$ and $|C| \geq |N'| \geq s$, contradicting that C is an atom.
- (F2) *Let $N \in \mathcal{L}$, $u \in U$ and $T := N(u) \cap Y$. Each anticomponent A of $G[T]$ satisfies $A \supseteq N$, or $A \cap N = \emptyset$, or $A \subsetneq N$.* This holds because A is a module of $G[Y]$ by hypothesis and N is strong. In the third case, if N is prime, Lemma 0.7 puts A inside a single child of N .
- (F3) *If $N \supsetneq N'$ are in $\mathcal{L}$, then $S(N) \cap N' = \emptyset$, and every vertex of $Y \setminus N'$ is complete or anticomplete to N' .* The first part is the argument of (F1), and the second holds because N' is a module.

Step 1: an antichain bound. We use the following counting fact. Let $2 \leq K_0 \leq K_1$ and $M \geq 0$, and let $\mathcal{A}$ be a family of sets, each of size at most M , such that for every integer $K \in [K_0, K_1]$ fewer than K members of $\mathcal{A}$ have size at least n/K^4 . Then

$$(1) \quad \sum_{A \in \mathcal{A}} |A| \leq (K_0 - 1)M + \frac{n}{K_0 - 1} + |\mathcal{A}| \frac{n}{K_1^4}.$$

Indeed, fewer than K_0 members have size at least n/K_0^4 , and each has size at most M . For $K_0 \leq K < K_1$, at most K members have size in $[n/(K+1)^4, n/K^4]$, contributing at most n/K^3 . The remaining members have size below n/K_1^4 . Finally $\sum_{K \geq K_0} K^{-3} \leq \sum_{K \geq K_0} (\frac{1}{K-1} - \frac{1}{K}) = \frac{1}{K_0-1}$.

Let K_0 be the least integer with $K_0^4 \geq k$ and put $a := K_0 - 1$. Then $a^4 < k$ and, since $k \geq 2^{16}$, $a \geq 15$. Moreover $K_0 \leq K_1$: from $a^8 < k^2 \leq 4/x$ and $a^7 \geq 8$ we get $8ax \leq a^8x < 4$, so $(a+1)x \leq 2ax < 1$.

Step 2: a heavy chain. Let $\mathcal{L}^* := \{N \in \mathcal{L} : |N| \geq t\}$ and let $\mathcal{M}$ be the set of minimal members of $\mathcal{L}^*$. They are pairwise disjoint modules of size at least $t \geq n/K_0^4$. So if $|\mathcal{M}| \geq K_0$, any K_0 of them form a pure $(K_0, n/K_0^4)$ -blockade by Lemma 0.6, which is (TL1). Assume $|\mathcal{M}| \leq a$.

Let $\mathcal{X}$ be the set of maximal members of $\mathcal{L} \setminus \mathcal{L}^*$. They are pairwise disjoint and have sizes in $[s, t)$, so $|\mathcal{X}| \leq n/s$. If for some $K \in [K_0, K_1]$ at least K of them have size at least n/K^4 , those K sets form a pure $(K, n/K^4)$ -blockade, which is (TL1). Otherwise (1) with $M = t$ gives

$$\sum_{A \in \mathcal{X}} |A| \leq a \frac{n}{k} + \frac{n}{a} + \frac{n}{s} \cdot \frac{n}{K_1^4} \leq \frac{n}{a^3} + \frac{n}{15} + \frac{4}{x^2} \cdot 16x^4n \leq \left(\frac{1}{3375} + \frac{1}{15} + 2^{-34} \right) n \leq \frac{n}{4},$$

using $a^4 < k$, $a \geq 15$, $K_1 \geq 1/(2x)$ and $x \leq 2^{-20}$. Every member of $\mathcal{L} \setminus \mathcal{L}^*$ lies inside a member of $\mathcal{X}$, and layers are disjoint, so the layers of members of $\mathcal{L} \setminus \mathcal{L}^*$ have total size at most $n/4$. By (F1),

$$\sum_{N \in \mathcal{L}^*} |S(N)| \geq \frac{3n}{4}.$$

Every $N \in \mathcal{L}^*$ contains a member of $\mathcal{M}$. Summing over $m \in \mathcal{M}$ and using $|\mathcal{M}| \leq a$, there is $m \in \mathcal{M}$ such that the family $\mathcal{C} := \{N \in \mathcal{L}^* : N \supseteq m\}$ satisfies

$$\sum_{N \in \mathcal{C}} |S(N)| \geq \frac{3n}{4a}.$$

Its members all contain m , so by laminarity $\mathcal{C}$ is a chain.

Step 3: a thick layer. Suppose $|S(N)| \geq t$ for some $N \in \mathcal{C}$, and put $S := S(N)$.

- $G[N]$ *disconnected*. The children of N are its components, so each component of $G[S]$ is an atom and has fewer than s vertices. Since $s = x^2 n/4 \leq xt \leq |S|/K_1$, Lemma 0.3 applied in the complement gives an anticomplete $(K_1, |S|/K_1^2)$ -blockade, and $|S|/K_1^2 \geq x^2 t = x^2 n/k$. This is (TL3).
- $\overline{G}[N]$ *disconnected*. The same argument with anticomponents gives a complete blockade: (TL3).
- N *prime*. Let $K' := \lfloor 1/(xk) \rfloor \geq 2$. Let $u \in U$ with $S \not\subseteq N(u)$ and $r := |N(u) \cap S| \geq x|S|/4$, and let $T := N(u) \cap Y$. Let A be an anticomponent of $G[T]$ meeting S . If $A \supseteq N$ then $S \subseteq N(u)$, which is excluded, and $A \cap N = \emptyset$ is impossible. So by (F2) A lies in a single child of N , and since it meets S , that child is an atom. Thus $N(u) \cap S$ is a union of anticomponents of $G[T]$, each inside an atom. These are the anticomponents of $G[N(u) \cap S]$, and each has fewer than s vertices. Now $r/K' \geq r x k \geq \frac{xt}{4} x k = s$, so Lemma 0.3 gives a complete $(K', r/K'^2)$ -blockade, and $r/K'^2 \geq \frac{xt}{4} (xk)^2 \geq x^3 n/4$. This is (TL4). If no such u exists, $Y' := S$ satisfies (TL5).

Step 4: all layers thin. Otherwise $|S(N)| < t$ for all $N \in \mathcal{C}$. Let L be the largest integer with $(8L)^4 \leq k^3$. Then $L \geq 8$ and $k^3 < (8L + 8)^4 \leq (9L)^4$, as required in (TL2). Since $(8La)^4 \leq k^3 a^4 < k^4$, we have $8La < k$ and so

$$2Lt = \frac{2Ln}{k} \leq \frac{3n}{4a} \leq \sum_{N \in \mathcal{C}} |S(N)|.$$

For $N \in \mathcal{C}$ let $P(N) := \sum_{N' \in \mathcal{C}, N' \supseteq N} |S(N')|$ be the layer mass from the top of the chain down to N . For $0 \leq j < L$ let the *chunk* Γ_j be the union of the layers $S(N)$ with $2jt < P(N) \leq 2(j+1)t$. For every $\theta \leq \sum_{N \in \mathcal{C}} |S(N)|$, the layers with $P(N) \leq \theta$ have total size in $(\theta - t, \theta]$, because each layer has fewer than t vertices. Taking $\theta = 2jt$ and $\theta = 2(j+1)t$ shows that every chunk has more than t vertices. If $j < j'$, every chain member contributing to $\Gamma_{j'}$ lies strictly inside every chain member contributing to Γ_j : its P -value is larger, and P can only grow down the chain.

Say that $v \in Y$ has *type 1* if v is complete to every $N' \in \mathcal{C}$ with $v \notin N'$. A vertex v without type 1 is anticomplete to every such N' . To see this, take a member $N'' \not\ni v$ to which v is not complete; by (F3) v is anticomplete to N'' . Any other $N' \not\ni v$ is comparable with N'' , and v is complete or anticomplete to it. If $N' \subseteq N''$, then v is anticomplete to N' . If $N' \supseteq N''$, then v has a non-neighbour in $N'' \subseteq N'$, so again v is anticomplete to N' . Now let $v \in S(N) \subseteq \Gamma_j$ and $w \in S(N') \subseteq \Gamma_{j'}$ with $j < j'$. Then $N' \subsetneq N$, so $v \notin N'$ by (F3), and v is adjacent to w exactly when v has type 1. Let $\Gamma'_j \subseteq \Gamma_j$ be the vertices of the majority type, so $|\Gamma'_j| \geq t/2 = n/(2k)$. Then $(\Gamma'_0, \dots, \Gamma'_{L-1})$ is a pure $(L, n/(2k))$ -blockade: (TL2). $\square$

Lemma 2.2 (Pair Lemma). *Let Y_1, Y_2 be sets of vertices and $\theta > 0$. Suppose every $w \in Y_1$ has $Y_2 \subseteq N(w)$ or $|N(w) \cap Y_2| < \theta|Y_2|$, and every $u \in Y_2$ has $Y_1 \subseteq N(u)$ or $|N(u) \cap Y_1| < \theta|Y_1|$. Then (Y_1, Y_2) is complete or weakly 2θ -sparse.*

Proof. Let $F := \{u \in Y_2 : Y_1 \subseteq N(u)\}$. If $|F| \geq \theta|Y_2|$, then every $w \in Y_1$ has at least $\theta|Y_2|$ neighbours in Y_2 , so $Y_2 \subseteq N(w)$ and the pair is complete. Otherwise $e(Y_1, Y_2) < |F||Y_1| + |Y_2|\theta|Y_1| < 2\theta|Y_1||Y_2|$. $\square$

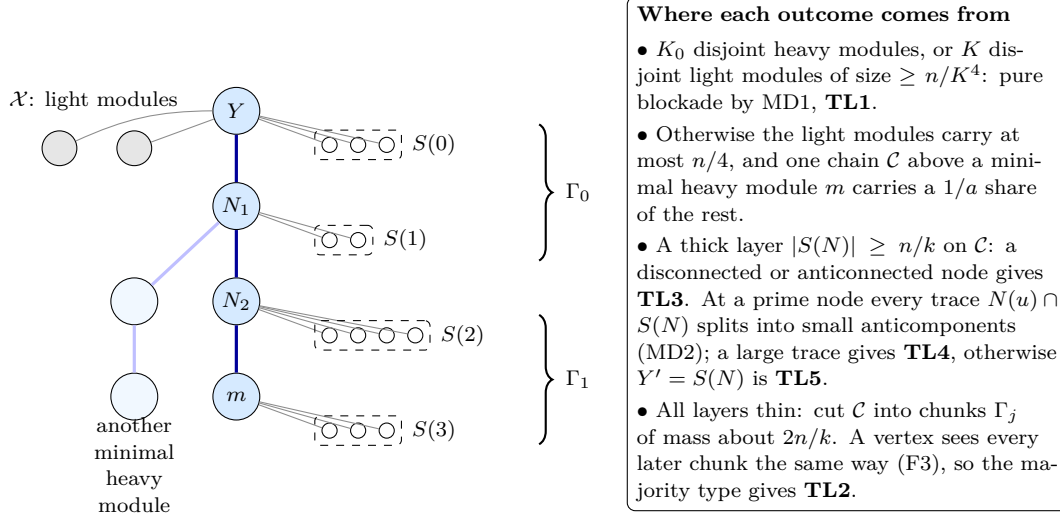


FIGURE 5. The proof of the Tooth Lemma. Circles are strong modules of $G[Y]$ with at least $s = x^2 n/4$ vertices. Blue: heavy modules, with at least $t = n/k$ vertices; the dark path is the chain $\mathcal{C}$ of heavy modules above one minimal heavy module m . Grey: light modules. Small white circles are atoms, and the dashed box $S(i)$ is the layer of the i -th member of the chain.

3. THE COMB LEMMA FOR $\overline{P_6}$

This section replaces [10, Lemma 5.2], the step where the proof for P_5 uses that the teeth of a comb are pure to each other. Lemma 3.1 has the same three outcomes, except that the blockade in outcome (ii) is only semisparsely.

Lemma 3.1. *Let $0 < x \leq y \leq 2^{-64}$, and let G be a y^3 -sparse $\overline{P_6}$ -free graph with $|G| \geq x^{-18}$. Then one of the following holds:*

- (i) G is $2y^4$ -sparse;
- (ii) for some integer K with $K^{16} \geq 1/y$ and $K \leq 1/x$ there is an x -semisparsely $(K, |G|/K^{100})$ -blockade;
- (iii) there are disjoint X, Y with $|X| \geq y^4|G|$ and $|Y| \geq (1 - 4y)|G|$ such that Y is x -sparse to X .

Proof. Step 1: a comb in a neighbourhood. This step follows [10, Claim 5.2.1]. Assume (i) and (iii) fail. Then some vertex v has $|N| \geq 2y^4|G|$, where $N := N(v)$. Note $|G| \geq x^{-18} \geq 2/y$. Let A' be the set of vertices $w \notin N \cup \{v\}$ with $|N(w) \cap N| \geq \frac{1}{2}y^2|N|$. Counting edges between A' and N and using y^3 -sparsity, $|A'| \cdot \frac{1}{2}y^2|N| \leq |N| \cdot y^3|G|$, so $|A'| \leq 2y|G|$. Let $A := V(G) \setminus (N \cup A' \cup \{v\})$. Since $|N| \leq y^3|G|$ and $1 \leq y|G|/2$, we get $|A| \geq (1 - 3y)|G|$.

Let $N' := \{w \in N : |N(w) \cap A| \leq x^2|A|\}$ and $B := N \setminus N'$. There are at most $x^2|A||N'|$ edges between A and N' , so at most $x|A|$ vertices of A have more than $x|N'|$ neighbours in N' . The remaining vertices of A , at least $(1 - x)|A| \geq (1 - 4y)|G|$ of them, form a set that is x -sparse to N' . Since (iii) fails, $|N'| < y^4|G| \leq |N|/2$, so $|B| \geq |N|/2 \geq y^4|G|$. Every vertex of A has fewer than $\frac{1}{2}y^2|N| \leq y^2|B|$ neighbours in B , and every vertex of B has more than $x^2|A|$ neighbours in A .

By Lemma 0.4 (with x^2 in place of x) there is $S \subseteq A$ with $|S| \leq x^{-2}$ such that at least $|B|/2$ vertices of B have a neighbour in S . Apply Theorem IV to (S, B) with $\Delta := y^2|B|$. Since $|B|/2 > 20y|B| = 20\sqrt{|B|\Delta}$, its first outcome fails, so there is an $(\ell, |B|/\ell^2)$ -comb $((a_i, B_i) : i \in [\ell])$ with

$a_i \in S$ and $B_i \subseteq B$. From $|B|/\ell^2 \leq |B_i| \leq |N(a_i) \cap B| \leq y^2|B|$ we get $\ell \geq 1/y \geq 2^{64}$, and since the a_i are distinct elements of S , $\ell \leq x^{-2}$. Each tooth satisfies $|B_i| \geq |B|/\ell^2 \geq y^4|G|/\ell^2$.

Step 2: the Tooth Lemma on every tooth. Let k be the least integer with $k^4 \geq \ell$. Then $(k-1)^4 < \ell$, so $2^{16} \leq k \leq \ell$ and $k-1 < \ell^{1/4} \leq x^{-1/2}$, whence $k \leq 2x^{-1/2}$. This also gives $k^2x \leq 4$ and $kx \leq 1$. Keep the teeth $B_1, \dots, B_k$ and put $n_i := |B_i|$. Since $y \geq x$ and $\ell \leq x^{-2}$,

$$n_i \geq \frac{y^4|G|}{\ell^2} \geq x^8|G| \geq x^{-10} \geq 16x^{-3}, \quad \text{and since } y \geq 1/\ell, \ell \leq k^4: \quad n_i \geq \frac{|G|}{\ell^6} \geq \frac{|G|}{k^{24}}.$$

Fix $i \in [k]$ and let $U_i := \bigcup_{j \in [k] \setminus \{i\}} B_j$. For $u \in U_i$ we have $v \not\sim a_i$ (as $a_i \in A$), $B_i \subseteq N(v) \cap N(a_i)$, $u \sim v$ (as $u \in N$), and $u \not\sim a_i$ (by the comb). By the module law (Lemma 1.1) with $Z = B_i$, every anticomponent of $N(u) \cap B_i$ is a module of $G[B_i]$. So the Tooth Lemma (Lemma 2.1) applies to $(Y, U) = (B_i, U_i)$ with this k ; its hypotheses $x \leq 2^{-20}$, $2^{16} \leq k \leq 2x^{-1/2}$ and $n_i \geq 16x^{-3}$ hold.

Step 3: one tooth gives (TL1)–(TL4). A pure or complete blockade is x -semisparse, and an anticomplete pair is weakly x -sparse. We check that the blockade has the length and width required in (ii).

- (TL1): $K^4 \geq k$ and $K \leq 1/x$. Then $K^{16} \geq k^4 \geq \ell \geq 1/y$, and the width is $n_i/K^4 \geq |G|/(k^{24}K^4) \geq |G|/K^{100}$ because $k^{24} \leq K^{96}$.
- (TL2): $(8L)^4 \leq k^3 \leq (9L)^4$. Then $L \geq k^{3/4}/9$, so $L^{16} \geq k^{12}/9^{16} \geq k^4 \geq 1/y$, and $L \leq k \leq 1/x$. The width is $n_i/(2k) \geq |G|/(2k^{25}) \geq |G|/L^{100}$, because $L^{100} \geq k^{75}/9^{100} \geq 2k^{25}$ for $k \geq 2^{16}$.
- (TL3): $K = \lfloor 1/x \rfloor \in [1/(2x), 1/x]$. Then $K^{16} \geq (2y)^{-16} \geq 1/y$, and the width is $x^2n_i/k \geq x^{11}|G| \geq (2x)^{100}|G| \geq |G|/K^{100}$, using $1/k \geq x$ and $n_i \geq x^8|G|$.
- (TL4): $K = \lfloor 1/(xk) \rfloor \geq 1/(2xk) \geq x^{-1/2}/4$, and $K \leq 1/x$. Then $K^{16} \geq x^{-8}/4^{16} \geq 1/y$, and the width is $x^3n_i/4 \geq x^{11}|G|/4 \geq |G|/K^{100}$.

In each case (ii) holds.

Step 4: every tooth gives (TL5). Then for each i there is $Y'_i \subseteq B_i$ with $|Y'_i| \geq n_i/k \geq |G|/k^{25}$ such that every $u \in U_i$ satisfies $Y'_i \subseteq N(u)$ or $|N(u) \cap Y'_i| < \frac{x}{4}|Y'_i|$. For $i \neq j$, every vertex of Y'_i lies in U_j and vice versa, so the Pair Lemma (Lemma 2.2) with $\theta = x/4$ shows that (Y'_i, Y'_j) is complete or weakly $\frac{x}{2}$ -sparse. Hence $(Y'_1, \dots, Y'_k)$ is an x -semisparse $(k, |G|/k^{100})$ -blockade, with $k^{16} \geq k^4 \geq \ell \geq 1/y$ and $k \leq 1/x$. This is (ii). $\square$

The comb lemma is the only place where the Tooth Lemma is used, and Step 4 is the reason the blockades of round one are semisparse rather than pure: two sets Y'_i, Y'_j can be sparse to each other on average while a few vertices of Y'_i are complete to Y'_j .

4. ROUND ONE: ITERATIVE SPARSIFICATION

This section follows [10, Lemmas 5.3–5.5], with Lemma 3.1 in place of their comb lemma. The only change is that the blockades produced are semisparse. Put $c := 2^{-8}$ and $y_0 := 2^{-64}$. Two elementary facts are used repeatedly: an x -sparse blockade is x -semisparse, and if $0 \leq t \leq \frac{1}{5}$ and $nt \leq 4$ then $(1-t)^n \geq e^{-5} > c$.

Lemma 4.1. *Let $0 < x \leq y \leq y_0$ and let G be a cy^3 -sparse $\overline{P_6}$ -free graph with $|G| \geq 2^8x^{-18}$. Then one of the following holds:*

- (a) *some $T \subseteq V(G)$ with $|T| \geq c|G|$ is $2y^4$ -sparse;*
- (b) *for some integer K with $K^{16} \geq 1/y$ and $K \leq 1/x$ there is an x -semisparse $(K, |G|/K^{102})$ -blockade;*
- (c) *there is an x -sparse $(1/y, y^6|G|)$ -blockade.*

Proof. Call $(B_0, \dots, B_n)$ a *chain* if the sets are disjoint, B_j is x -sparse to B_i for $i < j$, $|B_i| \geq y^6|G|$ for $i < n$, and $|B_n| \geq (1 - 4y)^n|G|$. The single set $(V(G))$ is a chain with $n = 0$, and $n \leq |G|$ for every chain, so we may take a chain with n maximal. If $n \geq 1/y$, then $(B_0, \dots, B_{n-1})$ is (c). Otherwise $n \cdot 4y \leq 4$ and $4y \leq \frac{1}{5}$, so $|B_n| \geq (1 - 4y)^n|G| \geq c|G|$. Hence $G[B_n]$ is y^3 -sparse, as each vertex has at most $cy^3|G| \leq y^3|B_n|$ neighbours, and $|B_n| \geq x^{-18}$. Apply Lemma 3.1 to $G[B_n]$. Outcome (i) is (a) with $T = B_n$. Outcome (ii) gives (b), because $K^{16} \geq 1/y \geq 2^{64}$ forces $K \geq 16$, so $|B_n|/K^{100} \geq c|G|/K^{100} \geq |G|/K^{102}$. In outcome (iii) we get disjoint $X, Y \subseteq B_n$ with $|X| \geq y^4|B_n| \geq y^4c|G| \geq y^6|G|$ and $|Y| \geq (1 - 4y)|B_n| \geq (1 - 4y)^{n+1}|G|$, with Y x -sparse to X . Subsets of B_n are x -sparse to every earlier block, so $(B_0, \dots, B_{n-1}, X, Y)$ is a longer chain, contradicting maximality. $\square$

Lemma 4.2. *Let $0 < x < y_0$ and let G be a cy_0^3 -sparse $\overline{P}_6$ -free graph with $|G| \geq 2^{16}x^{-19}$. Then one of the following holds:*

- (1) *for some integer K with $16 \leq K \leq 1/x$ there is an x -semisparse $(K, |G|/K^{118})$ -blockade;*
- (2) *for some $y \in [x, y_0]$ there is an x -sparse $(1/y, y^7|G|)$ -blockade.*

Proof. For $j \geq 0$ let $y_j := y_0 c^j$. Call j *admissible* if $xc \leq y_j$ and some $F \subseteq V(G)$ with $|F| \geq y_j|G|$ is cy_j^3 -sparse. Then $j = 0$ is admissible with $F = V(G)$, and admissible j are bounded because $y_j \rightarrow 0$. Let j be the largest admissible index, $y := y_j$, and F a witness.

Case $y \geq x$. Then $|F| \geq y|G| \geq x|G| \geq 2^8x^{-18}$, so Lemma 4.1 applies to $G[F]$. In outcome (a), $T \subseteq F$ has $|T| \geq c|F| \geq y_{j+1}|G|$ and is $2y^4$ -sparse, hence cy_{j+1}^3 -sparse because $2y^4 \leq y^3c^4$. Also $xc \leq yc = y_{j+1}$. So $j+1$ is admissible, a contradiction. Outcome (b) gives (1): $K^{16} \geq 1/y \geq 2^{64}$ gives $K \geq 16$, and $|F|/K^{102} \geq y|G|/K^{102} \geq |G|/K^{118}$ because $yK^{16} \geq 1$. Outcome (c) is an x -sparse $(1/y, y^6|F|)$ -blockade with $y^6|F| \geq y^7|G|$, which is (2).

Case $y < x$. Then F is x^3 -sparse and $|F| \geq y|G| \geq xc|G|$, so $|F| \geq 4/x$. Let $m := \lceil 1/x \rceil \leq 2/x$ and cut F into m disjoint parts of size $\lfloor |F|/m \rfloor \geq x|F|/4$. Every vertex has at most $x^3|F| \leq x \cdot x|F|/4$ neighbours in each part, so the parts form an x -sparse $(1/x, x|F|/4)$ -blockade, and $x|F|/4 \geq x^7|G|$. This is (2) with $y = x$. $\square$

The next lemma is round one in its final form. It is the only place where the constant d of the proof is fixed.

Lemma 4.3. *There is an integer $d \geq 200$ such that for every $x \in (0, 2^{-d})$ and every $\overline{P}_6$ -free graph G with $|G| \geq x^{-d}$ there are a real $k \in [2, 1/x]$ and an x -semisparse $(k, |G|/k^d)$ -blockade.*

Proof. Let $\delta := \delta(2^{-200})$ from Theorem I, let L be an integer with $2^L\delta \geq 1$, and put $d := L + 200$. Let $x \in (0, 2^{-d})$ and $|G| \geq x^{-d}$. By Theorem I there is F with $|F| \geq \delta|G|$ that is 2^{-200} -restricted. Note $2^{200} \leq \delta 2^d \leq \delta/x$.

Case $\overline{G}[F]$ is 2^{-200} -sparse. Then $\overline{G}[F]$ is 360^{-2} -sparse and has no induced P_6 . Theorem II with $y = 1/360$ gives an anticomplete $(360, \lfloor 360^{-2}|F| \rfloor)$ -blockade in $\overline{G}[F]$, that is, a complete one in G . Since $360^{-2}\delta|G| \geq 360^{-2}\delta 2^d \geq 2$, the width is at least $360^{-2}|F|/2 \geq |G|/2^d$. So $k = 2$ works.

Case $G[F]$ is 2^{-200} -sparse. Note $2^{-200} = cy_0^3$ and $x < y_0$. Also $|F| \geq \delta x^{-d} \geq 2^{16}x^{-19}$. Apply Lemma 4.2 to $G[F]$. In outcome (1), take $k = K \in [16, 1/x]$: the width is $|F|/K^{118} \geq \delta|G|/K^{118} \geq |G|/K^d$ because $K^{L+82} \geq 2^L \geq 1/\delta$. In outcome (2), take $k := \lfloor 1/y \rfloor$. Then $2^{64} \leq k \leq 1/y \leq 1/x$ and $y \geq 1/(2k)$, so the width is $y^7|F| \geq (2k)^{-7}\delta|G| \geq |G|/k^d$. An x -sparse blockade is x -semisparse. $\square$

5. THE LAYOUT THEOREM AND NICENESS

The layout theorem of Nguyen, Scott and Seymour [10, Theorem 6.1] turns “every large induced subgraph has a short blockade of polynomial width” into “the graph has a long blockade

of polynomial width”. As stated there, its input blockades are pure. Its proof only counts *wrong* pairs, namely edges between blocks that are declared anticomplete, and a weakly sparse pair of blocks contributes few of them. So the same proof accepts semisparsed input. We give it in full.

Theorem 5.1. *Let $\varepsilon \in (0, 1)$, let $d \geq 1$ be an integer, put $x := \varepsilon^{5d}$, and let G be a graph. Suppose every induced subgraph F with $|F| \geq \varepsilon^d |G|$ has, for some real $k \in [2, 1/x]$, an x -semisparsed $(k, |F|/k^d)$ -blockade. Then G has an ε^d -semisparsed $(1/\varepsilon, x^{2d}|G|)$ -blockade.*

Proof. If G has no vertices, take $\lceil 1/\varepsilon \rceil$ empty blocks. Otherwise, a *layout* is a family $\mathcal{F}$ of pairwise disjoint nonempty sets $A \subseteq V(G)$ together with a set $\mathcal{G}$ of pairs of distinct members of $\mathcal{F}$ declared *complete*, such that every declared pair is complete in G . An ordered pair (u, w) of distinct vertices is *decided* if no member of $\mathcal{F}$ contains both, and *wrong* if $uw \in E(G)$, $u \in A$, $w \in B$ for distinct $A, B \in \mathcal{F}$, and $\{A, B\} \notin \mathcal{G}$. A layout is *good* if

- (L1) $|A| \geq \varepsilon^{2d} |G|$ for every $A \in \mathcal{F}$;
- (L2) $\sum_{A \in \mathcal{F}} |A|^{1/d} \geq |G|^{1/d}$;
- (L3) the number of wrong pairs is at most x times the number of decided pairs.

The layout $\mathcal{F} = \{V(G)\}$, $\mathcal{G} = \emptyset$ is good, and $|\mathcal{F}| \leq |G|$ always. Choose a good layout with $|\mathcal{F}|$ maximum.

Claim 1: if $|\mathcal{F}| \geq 1/\varepsilon$, the members of $\mathcal{F}$ form the required blockade. For $A, B \in \mathcal{F}$ with $\{A, B\} \in \mathcal{G}$ the pair is complete. Otherwise every edge between A and B is wrong, so

$$e(A, B) \leq x \cdot \#\text{decided} \leq x|G|^2 \leq x\varepsilon^{-4d}|A||B| = \varepsilon^d|A||B|$$

by (L1). Also $|A| \geq \varepsilon^{2d}|G| \geq x^{2d}|G|$.

So assume $|\mathcal{F}| < 1/\varepsilon$. Let A be a largest member of $\mathcal{F}$. By (L2), $|G|^{1/d} \leq |\mathcal{F}| |A|^{1/d}$, so $|A| \geq |\mathcal{F}|^{-d} |G| \geq \varepsilon^d |G|$. By hypothesis $G[A]$ has an x -semisparsed blockade $(B_1, \dots, B_m)$ with $m \geq k$, $k \in [2, 1/x]$ and $|B_p| \geq |A|/k^d > 0$.

Claim 2: $k \geq 1/\varepsilon$. Suppose $k < 1/\varepsilon$. Replace A in $\mathcal{F}$ by $B_1, \dots, B_m$. Declare $\{B_p, C\}$ complete when $\{A, C\} \in \mathcal{G}$, and $\{B_p, B_q\}$ complete when B_p is complete to B_q . This is a layout. We check that it is good. (L1): $|B_p| \geq |A|/k^d \geq \varepsilon^d |A| \geq \varepsilon^{2d} |G|$. (L2): $\sum_p |B_p|^{1/d} \geq m(|A|/k^d)^{1/d} = \frac{m}{k} |A|^{1/d} \geq |A|^{1/d}$. (L3): Let R be the set of ordered pairs with ends in distinct blocks B_p, B_q . Refining A keeps every decided pair decided, and the pairs in R were undecided before, so the new decided pairs include the old ones and R , disjointly. A new wrong pair either was already wrong, or lies in R and is an edge between blocks B_p, B_q that are not complete to each other. Pairs with an end in $A \setminus \bigcup_p B_p$ are never wrong, because that end lies in no member of the new layout. Blocks that are not complete are weakly x -sparse, so the new wrong pairs number at most $x \cdot \#\text{old decided} + x|R| \leq x \cdot \#\text{new decided}$. Since $m \geq 2$, the new layout has more members than $\mathcal{F}$, contradicting maximality. This proves Claim 2.

So $k \geq 1/\varepsilon$, and $(B_1, \dots, B_m)$ has at least $1/\varepsilon$ blocks, each of size at least $|A|/k^d \geq x^d |A| \geq x^d \varepsilon^d |G| \geq x^{2d} |G|$. Every two blocks are complete or weakly x -sparse, hence weakly ε^d -sparse. $\square$

Lemma 5.2 (P_6 is nice). *Let d be as in Lemma 4.3. For every $\varepsilon \in (0, \frac{1}{2})$ and every $\overline{P}_6$ -free graph G with $|G| \geq \varepsilon^{-10d^2}$, G has an ε^d -semisparsed $(1/\varepsilon, \varepsilon^{10d^2}|G|)$ -blockade.*

Proof. Put $x := \varepsilon^{5d}$; then $x < 2^{-d}$. If $|F| \geq \varepsilon^d |G|$ then $|F| \geq \varepsilon^{d-10d^2} \geq \varepsilon^{-5d^2} = x^{-d}$, so Lemma 4.3 gives the hypothesis of Theorem 5.1 for F . The conclusion is the claim, since $x^{2d} = \varepsilon^{10d^2}$. $\square$

6. ROUND TWO: MIXED VERTICES AND THE HOUSE-FREE PATTERN

This section follows [10, Lemmas 7.1 and 7.2]. The new ingredient is the claim in Step 4 of the proof of Lemma 6.1: where the proof for P_5 uses the house to show that an outside vertex cannot be mixed on two complete blocks, we use Corollary 1.4.

Throughout, $d \geq 200$ is the integer of Lemma 4.3, $\kappa > 0$ is the exponent of Theorem III, M is an integer with $M \geq 5/\kappa + 4$, and

$$N := 10d^2, \quad A_1 := M(N + 3).$$

Lemma 6.1. *Let $y \in (0, \frac{1}{8}]$ and let G be a y -sparse $\overline{P}_6$ -free graph. Then one of the following holds:*

- (a) *some $T \subseteq V(G)$ with $|T| \geq y^{A_1}|G|$ is y^4 -sparse;*
- (b) *G has a complete $(1/y, y^{A_1}|G|)$ -blockade;*
- (c) *there are disjoint X, Y with $|X| \geq y^{A_1}|G|$ and $|Y| \geq (1 - 4y)|G|$, X anticomplete to Y .*

Proof. Assume (a) and (b) fail. Put $\varepsilon := y^M$. Since $y \leq \frac{1}{8}$ and $M \geq 5$,

$$\varepsilon \leq 2^{-15}, \quad \varepsilon \leq y^5, \quad 8\varepsilon \leq y^2, \quad y^{A_1} = \varepsilon^{N+3}.$$

If $|G| \leq \varepsilon^{-(N+3)}$, a single vertex gives (a). So $|G| > \varepsilon^{-(N+3)}$. Let

$$\ell := \lceil 1/\varepsilon \rceil, \quad m := \lceil \varepsilon^N |G| \rceil, \quad b := \lceil \varepsilon^2 m \rceil.$$

Then $1/\varepsilon \leq \ell \leq 2/\varepsilon$, $\varepsilon^N |G| \leq m \leq 2\varepsilon^N |G|$ and $m > \varepsilon^{-3}$, so $1 \leq \varepsilon^2 m \leq b \leq 2\varepsilon^2 m$. Consequently $b \leq m/(2\ell)$ and $(\ell - 1)\varepsilon^2 m \leq m/2$, because $4\varepsilon^2 \ell \leq 8\varepsilon \leq 1$.

Step 1: a semisparse blockade. By Lemma 5.2 (with $\varepsilon < \frac{1}{2}$ and $|G| \geq \varepsilon^{-N}$) there are disjoint $A_1, \dots, A_\ell$ with $|A_i| \geq \varepsilon^N |G|$, hence $|A_i| \geq m$, such that every two of them are complete or weakly ε^d -sparse. Below, a pair $p \neq q$ is *non-complete* if A_p is not complete to A_q .

Step 2: equal sizes. Put $K_1 := 2\ell\varepsilon^d$ and $K_2 := \varepsilon^{d-3}$; then $2\ell K_1 = 4\ell^2\varepsilon^d \leq K_2$. For $i = 1, \dots, \ell$ in turn we choose $X_i \subseteq A_i$ with $|X_i| = m$ such that, for all non-complete pairs among the indices chosen so far,

(P1) $e(X_p, A_q) \leq K_1 m |A_q|$ whenever $p < q$ (here q need not be chosen yet);

(P2) $e(X_{p'}, X_p) \leq K_2 m^2$ whenever $p' < p$.

Suppose $X_1, \dots, X_{i-1}$ are chosen. Let $n := \lfloor |A_i|/m \rfloor \geq 1$ and let $P_1, \dots, P_n$ be disjoint subsets of A_i of size m ; note $|A_i| \leq 2nm$. For each $q \neq i$ with (q, i) non-complete, give P_j the score $e(X_q, P_j)/(2K_1 m^2)$ if $q < i$, and $e(P_j, A_q)/(2\varepsilon^d m |A_q|)$ if $q > i$. For fixed q the scores of $P_1, \dots, P_n$ add up to at most n : if $q < i$ this is (P1) for (q, i) , since $\sum_j e(X_q, P_j) \leq e(X_q, A_i) \leq K_1 m |A_i| \leq 2K_1 n m^2$; if $q > i$ it is weak sparsity, since $\sum_j e(P_j, A_q) \leq \varepsilon^d |A_i| |A_q| \leq 2\varepsilon^d n m |A_q|$. Summing over at most ℓ indices q , some P_j has total score at most ℓ , and we put $X_i := P_j$. Then every single score of X_i is at most ℓ . This gives $e(X_i, A_q) \leq 2\ell\varepsilon^d m |A_q| = K_1 m |A_q|$ for $q > i$, and $e(X_q, X_i) \leq 2\ell K_1 m^2 \leq K_2 m^2$ for $q < i$.

Step 3: anticonnected blocks. Put $\theta_1 := \varepsilon^{d-5}$ and $\theta_2 := \varepsilon^{d-7}$. Then $K_2 = \varepsilon^2 \theta_1$, $\theta_1 = \varepsilon^2 \theta_2$ and $\theta_1 m \leq \theta_2 b$. For $i = 1, \dots, \ell$ in turn we choose an anticonnected $B_i \subseteq X_i$ with $|B_i| = b$ such that, for all non-complete pairs,

(Q1) B_p is θ_1 -sparse to X_q whenever $p < q$;

(Q2) B_p is θ_2 -sparse to $B_{p'}$ and $B_{p'}$ is θ_2 -sparse to B_p whenever $p' < p$.

Suppose $B_1, \dots, B_{i-1}$ are chosen. For $q \neq i$ with (q, i) non-complete let

$$C_q := \begin{cases} \{w \in X_i : |N(w) \cap B_q| > \theta_2 b\} & (q < i), \\ \{w \in X_i : |N(w) \cap X_q| > \theta_1 m\} & (q > i). \end{cases}$$

If $q < i$, (Q1) gives $e(B_q, X_i) \leq b\theta_1 m$, so $|C_q| \leq \theta_1 m / \theta_2 = \varepsilon^2 m$. If $q > i$, (P2) gives $e(X_i, X_q) \leq K_2 m^2$, so $|C_q| \leq K_2 m / \theta_1 = \varepsilon^2 m$. Let $D := X_i \setminus \bigcup_q C_q$. Then $|D| \geq m - (\ell - 1)\varepsilon^2 m \geq m/2$. If every anticomponent of D has fewer than $|D|/\ell$ vertices, Lemma 0.3 gives a complete $(\ell, |D|/\ell^2)$ -blockade. Here $\ell \geq 1/\varepsilon \geq 1/y$ and $|D|/\ell^2 \geq \frac{m}{2} \cdot \frac{\varepsilon^2}{4} \geq \varepsilon^{N+3}|G|$, so this is (b), which we excluded. Hence some anticomponent of D has at least $|D|/\ell \geq m/(2\ell) \geq b$ vertices, and by Lemma 0.5 it contains an anticonnected set B_i of exactly b vertices. Since B_i avoids C_q for $q > i$, (Q1) holds for B_i . Since B_i avoids C_q for $q < i$, B_i is θ_2 -sparse to B_q . Conversely, by (Q1) every vertex of B_q has at most $\theta_1 m \leq \theta_2 b$ neighbours in $X_i \supseteq B_i$, so B_q is θ_2 -sparse to B_i .

Step 4: a block system. The sets $B_1, \dots, B_\ell$ are disjoint, anticonnected and of size b . Two of them are complete if the corresponding A 's are, and otherwise θ_2 -sparse to each other in both directions. Since $\theta_2 = \varepsilon^{d-7} \leq \varepsilon < \frac{1}{2}$, they form a block system in the sense of Section 1, with $W = b$ and $x = \theta_2$. Let $U := \bigcup_i B_i$ and $R_0 := V(G) \setminus U$. Since $|U| = \ell b \leq \frac{2}{\varepsilon} \cdot 2\varepsilon^2 m \leq 8\varepsilon \cdot \varepsilon^N |G| \leq y^2 |G|$, we have $|R_0| \geq (1 - y^2)|G|$.

Claim: every $v \in R_0$ has $|I(v)| < y\ell$. Suppose $|I(v)| \geq y\ell$. Then $|I(v)| \geq y/\varepsilon = y^{-(M-1)}$, and since $(M-1)\kappa \geq 5$,

$$|I(v)|^\kappa \geq y^{-(M-1)\kappa} \geq y^{-5}.$$

By Corollary 1.4 there is $R \subseteq I(v)$ with $r := |R| \geq y^{-5}$ whose blocks are pairwise complete or pairwise non-complete. Let $T := \bigcup_{i \in R} B_i$, so $|T| = rb$, and note $b \geq \varepsilon^2 m \geq \varepsilon^{N+2}|G| \geq y^{A_1}|G|$. If the blocks are pairwise non-complete, they are pairwise θ_2 -sparse, and a vertex of T has fewer than $b \leq y^5|T|$ neighbours in its own block and at most $r\theta_2 b \leq y^5|T|$ in the others, because $\theta_2 \leq \varepsilon \leq y^5$. So T is $2y^5$ -sparse, hence y^4 -sparse, and $|T| \geq y^{A_1}|G|$: this is (a). If the blocks are pairwise complete, they form a complete (r, b) -blockade with $r \geq y^{-5} \geq 1/y$: this is (b). Both were excluded, which proves the claim.

Step 5: the anticomplete pair. By the claim, $\sum_{v \in R_0} |I(v)| < y\ell|R_0|$, so some index i belongs to $I(v)$ for fewer than $y|R_0|$ vertices $v \in R_0$. Since G is y -sparse, $e(V(G), B_i) \leq by|G|$, so at most $2y|G|$ vertices have at least $b/2$ neighbours in B_i . Let Y be the set of $v \in R_0$ with no neighbour in B_i . A vertex of $R_0 \setminus Y$ has between 1 and $b/2$ neighbours in B_i , so $i \in I(v)$, or it has at least $b/2$. Therefore

$$|Y| \geq |R_0| - y|R_0| - 2y|G| \geq (1 - y)(1 - y^2)|G| - 2y|G| \geq (1 - 4y)|G|.$$

With $X := B_i$ we have $|X| = b \geq y^{A_1}|G|$, and X is anticomplete to Y : this is (c). $\square$

Lemma 6.2. *Suppose $A_1 = 2k$ for an integer k . Let $y \in (0, 2^{-16}]$ and let G be a y -sparse $\overline{P_6}$ -free graph. Then one of the following holds:*

- (i) *some $T \subseteq V(G)$ with $|T| \geq y^{k+1}|G|$ is y^2 -sparse;*
- (ii) *G has a complete or anticomplete $(y^{-1/2}, y^{k+1}|G|)$ -blockade.*

Proof. Let $s := y^{1/2} \leq 2^{-8}$, so $256y \leq s$ and $s^{A_1} = y^k$. Call $(B_0, \dots, B_n)$ a chain if the sets are disjoint and pairwise anticomplete, $|B_i| \geq y^{k+1}|G|$ for $i < n$, and $|B_n| \geq (1 - 4s)^n|G|$. Take a chain with n maximal. If $n \geq 1/s$, then $(B_0, \dots, B_{n-1})$ gives (ii). Otherwise $|B_n| \geq c|G|$ with $c = 2^{-8}$, as in Lemma 4.1. Each vertex of B_n has at most $y|G| \leq 256y|B_n| \leq s|B_n|$ neighbours, so $G[B_n]$ is s -sparse, and Lemma 6.1 applies with $s \leq \frac{1}{8}$. Outcome (a) gives $T \subseteq B_n$ with $|T| \geq y^k|B_n| \geq y^k c|G| \geq y^{k+1}|G|$ that is $s^4 = y^2$ -sparse: this is (i). Outcome (b) gives a complete $(1/s, y^k|B_n|)$ -blockade: this is (ii). Outcome (c) gives an anticomplete pair $X, Y \subseteq B_n$ with $|X| \geq y^{k+1}|G|$ and $|Y| \geq (1 - 4s)^{n+1}|G|$, and $(B_0, \dots, B_{n-1}, X, Y)$ is a longer chain, which is impossible. $\square$

7. CRUX (C) FOR $\overline{P_6}$ -FREE GRAPHS

Nguyen, Scott and Seymour reduce the Erdős–Hajnal property to a statement about sparse graphs, their “crux” [10, Section 7]. We use the following instance of it, in which the sparsity exponent is 2, the length exponent is $\frac{1}{2}$ and the threshold is 2^{-16} .

Definition 7.1 (Crux (C) with exponent a). Let $a \geq 1$ be an integer. *Crux (C) holds with exponent a* if for every $\overline{P_6}$ -free graph G and every $y \in (0, 2^{-16}]$ such that G is y -sparse, one of the following holds:

- (i) some $T \subseteq V(G)$ with $|T| \geq y^a |G|$ is y^2 -sparse;
- (ii) G has a complete or anticomplete $(y^{-1/2}, y^a |G|)$ -blockade.

Corollary 7.2. Let d be the integer of Lemma 4.3 and κ the exponent of Theorem III, and put $M' := \lceil 5/\kappa + 4 \rceil$. Then *Crux (C) holds with exponent $a := M'(10d^2 + 3) + 1$* .

Proof. Apply Lemma 6.1 with the even integer $M := 2M' \geq 5/\kappa + 4$. Then $A_1 = M(10d^2 + 3) = 2k$ with $k = M'(10d^2 + 3)$, and Lemma 6.2 gives Crux (C) with exponent $k + 1 = a$. $\square$

Section 8 shows that Crux (C), together with Theorems I and II, implies the polynomial Rödl property and Theorem A. Table 1 collects all the constants of the proof. Every constant is determined by the constants of the imported theorems. The values are those of the Lean proof.

constant	definition	fixed in
δ_1	$\delta(2^{-200})$ from Rödl’s theorem (Theorem I)	Lemma 4.3
L_1	an integer with $2^{L_1} \delta_1 \geq 1$	Lemma 4.3
d	$L_1 + 200$ (so $d \geq 200$)	Lemma 4.3
κ	the exponent of Theorem III; necessarily $\kappa \leq 1$	Theorem III
M', M	$M' = \lceil 5/\kappa + 4 \rceil \geq 9$, $M = 2M'$	Corollary 7.2
A_1	$M(10d^2 + 3)$	Lemma 6.1
a	$A_1/2 + 1 = M'(10d^2 + 3) + 1$	Corollary 7.2
δ_2	$\delta(2^{-17})$ from Rödl’s theorem	Lemma 8.1
t	$t_0 + a + 1$, where $2^{-17t_0} < \delta_2$	Lemma 8.1
L_2	an integer with $2^{L_2} \cdot 360^{-2} \delta_2 > 2$	Lemma 8.1
A	$2(a + t) + L_2$	Lemma 8.1
τ	$1/((3A + 1)(3A + 2))$	Theorem 8.3

TABLE 1. The constants. The only unknowns are δ_1, δ_2 from Rödl’s theorem and κ from the Erdős–Hajnal theorem for P_5 . Even with the most favourable values, $d \geq 200$ and $M' \geq 9$ give $a \geq 3.6 \cdot 10^6$, $A \geq 4a$ and $1/\tau > 9A^2 > 10^{15}$.

8. FROM CRUX (C) TO THE ERDŐS–HAJNAL PROPERTY

This section follows [10, Section 7], where the analogous reduction is carried out for P_5 . Lemma 8.2 does not depend on the forbidden graph. We write the proof of Lemma 8.2 as an induction that produces two families of blocks, which is the form used in the formalization. Throughout, a is the exponent of Corollary 7.2.

Lemma 8.1. *There is an integer $A \geq 1$ such that for every $x \in (0, \frac{1}{2})$ and every $\overline{P_6}$ -free graph G , either some induced subgraph F with $|F| \geq x^A |G|$ is x -restricted, or G has a complete or anticomplete $(k, |G|/k^A)$ -blockade for some integer k with $2 \leq k \leq 1/x$.*

Proof. Let $c_0 := 2^{-17}$ and $\eta := 360^{-2}$, and let δ_2, t, L_2, A be as in Table 1. Thus $c_0^t \leq \delta_2$, $\eta\delta_2 2^{L_2} > 2$, $t \geq a$, and $A \geq 2(a+t)$, $A \geq L_2$. If $|G| < x^{-A}$, a single vertex is an x -restricted subgraph with $1 \geq x^A|G|$. So assume $|G| \geq x^{-A} \geq 2^A$. By Theorem I there is F_0 with $|F_0| \geq \delta_2|G|$ that is c_0 -restricted.

Case $x \geq c_0$. Then F_0 is x -restricted, and $|F_0| \geq \delta_2|G| \geq 2^{-A}|G| \geq x^A|G|$, because $\delta_2 2^A \geq \delta_2 2^{L_2} \geq 1$.

Case $x < c_0$ and $\overline{G}[F_0]$ c_0 -sparse. Then $\overline{G}[F_0]$ is η -sparse and has no induced P_6 , so Theorem II with $y = 1/360$ gives a complete $(360, \lfloor \eta|F_0| \rfloor)$ -blockade in G . Since $\eta|F_0| \geq \eta\delta_2 2^A \geq 2$, its width is at least $\eta\delta_2|G|/2 \geq |G|/2^A$. This is the second outcome with $k = 2 \leq 1/x$.

Case $x < c_0$ and $G[F_0]$ c_0 -sparse. For $j \geq 0$ let $y_j := c_0^{2^j}$, so $y_{j+1} = y_j^2$. Call j admissible if $x^2 \leq y_j$ and some F with $|F| \geq y_j^t|G|$ is y_j -sparse. Then $j = 0$ is admissible, witnessed by F_0 , and admissible j are bounded. Let j be the largest admissible index, $y := y_j \leq c_0$, and F a witness.

If $x \leq y$, apply Crux (C) to $G[F]$. In outcome (i) there is a y^2 -sparse $T \subseteq F$ with $|T| \geq y^a|F| \geq y^{a+t}|G| \geq (y^2)^t|G|$, and $x^2 \leq y^2$, so $j+1$ is admissible, which is impossible. In outcome (ii) there is a complete or anticomplete $(y^{-1/2}, y^a|F|)$ -blockade. Let $k := \lceil y^{-1/2} \rceil$. Then $k \geq 2$, $k \leq 2y^{-1/2} \leq 2x^{-1/2} \leq 1/x$, and $k^{-2} \leq y$, so the width is at least $y^{a+t}|G| \geq k^{-2(a+t)}|G| \geq |G|/k^A$. This is the second outcome.

If $y < x$, then F is x -sparse and $|F| \geq y^t|G| \geq x^{2t}|G| \geq x^A|G|$. This is the first outcome. $\square$

Lemma 8.2 (cf. [10, Theorem 7.4]). *Let $\varepsilon \in (0, 1)$, let $A \geq 1$ be an integer, and let G be a graph such that every induced subgraph F with $|F| \geq \varepsilon^{2A}|G|$ has a complete or anticomplete $(k, |F|/k^A)$ -blockade for some integer $k \in [2, 1/\varepsilon]$. Then G has an ε -restricted induced subgraph with at least $\varepsilon^{3A}|G|$ vertices.*

Proof. We may assume $|G| \geq 1$. Put $T_0 := \varepsilon^{2A}|G|$ and $w := \varepsilon^{3A}|G| = \varepsilon^A T_0$. We show by induction on $|F|$: for every $F \subseteq V(G)$ with $|F| \geq w$ there are families $\mathcal{P}, \mathcal{Q}$ of pairwise disjoint subsets of F , each of size at least w , such that the members of $\mathcal{P}$ are pairwise anticomplete, the members of $\mathcal{Q}$ are pairwise complete, and

$$|F| \leq (|\mathcal{P}||\mathcal{Q}|)^A T_0.$$

If $|F| < T_0$, take $\mathcal{P} = \mathcal{Q} = \{F\}$. Otherwise the hypothesis gives a complete or anticomplete blockade $(B_1, \dots, B_m)$ in $G[F]$ with $m \geq k \geq 2$, $k\varepsilon \leq 1$ and $|B_i| \geq |F|/k^A \geq \varepsilon^A|F| \geq \varepsilon^A T_0 = w$. Each B_i is smaller than F , because another block is nonempty. By induction each B_i has families $\mathcal{P}_i, \mathcal{Q}_i$ with $|B_i| \leq (|\mathcal{P}_i||\mathcal{Q}_i|)^A T_0$. Let $\mu := \min_i |\mathcal{P}_i||\mathcal{Q}_i|$, attained at i_0 , so $\sum_i |\mathcal{P}_i||\mathcal{Q}_i| \geq m\mu \geq k\mu$. If the blockade is anticomplete, let $\mathcal{P} := \bigcup_i \mathcal{P}_i$, which is pairwise anticomplete because different blocks are anticomplete, and let $\mathcal{Q}$ be a largest $\mathcal{Q}_i$. If it is complete, let $\mathcal{Q} := \bigcup_i \mathcal{Q}_i$ and $\mathcal{P}$ a largest $\mathcal{P}_i$. In both cases $|\mathcal{P}||\mathcal{Q}| \geq \sum_i |\mathcal{P}_i||\mathcal{Q}_i| \geq k\mu$, and

$$|F| \leq k^A |B_{i_0}| \leq k^A \mu^A T_0 \leq (|\mathcal{P}||\mathcal{Q}|)^A T_0.$$

Apply this to $F = V(G)$: $|G| \leq (|\mathcal{P}||\mathcal{Q}|)^A \varepsilon^{2A}|G|$ gives $|\mathcal{P}||\mathcal{Q}| \geq \varepsilon^{-2}$, so one of the families has at least $1/\varepsilon$ members. Suppose it is $\mathcal{P}$, with $r \geq 1/\varepsilon$ members. Choose in each member a subset of exactly $\lceil w \rceil$ vertices, and let T be their union. Then $|T| = r\lceil w \rceil \geq w$, and each vertex of T has all its neighbours in T inside its own piece, so it has fewer than $\lceil w \rceil = |T|/r \leq \varepsilon|T|$ of them. So T is ε -sparse. If the family is $\mathcal{Q}$, the same argument in $\overline{G}$ shows that $\overline{G}[T]$ is ε -sparse. $\square$

Theorem 8.3 (polynomial Rödl property). *Let A be as in Lemma 8.1. For every $\varepsilon \in (0, \frac{1}{2})$, every $\overline{P}_6$ -free graph G has an ε -restricted induced subgraph with at least $\varepsilon^{3A}|G|$ vertices. The same holds for every P_6 -free graph.*

Proof. If some F with $|F| \geq \varepsilon^{2A}|G|$ has an ε -restricted induced subgraph with at least $\varepsilon^A|F| \geq \varepsilon^{3A}|G|$ vertices, we are done. Otherwise Lemma 8.1, applied with $x = \varepsilon$ to each such F , gives the hypothesis of Lemma 8.2, which gives the claim. The statement for P_6 -free graphs follows by complementation, since the definition of restricted is symmetric. $\square$

Proof of Theorem A. Let G be P_6 -free with n vertices; then $\overline{G}$ is $\overline{P_6}$ -free. Let $\tau := 1/((3A+1)(3A+2))$ and $u := n^\tau$. If $u < 2$, then either $n \leq 1$ and $V(G)$ is a clique of size $n \geq u$, or two vertices form a clique or a stable set of size $2 > u$. Otherwise let $z := u^{3A+2} = n^{1/(3A+1)}$. Then $z \geq 2u \geq 4$, so $\varepsilon := 1/z < \frac{1}{2}$. Theorem 8.3 applied to $\overline{G}$ gives T with $|T| \geq \varepsilon^{3A}n = z$ that is ε -restricted. A graph on T with maximum degree at most D has a stable set of size at least $|T|/(D+1)$, by the greedy algorithm. If $\overline{G}[T]$ is ε -sparse, it has a stable set, which is a clique of G , of size at least $|T|/(\varepsilon|T|+1) \geq |T|/(2\varepsilon|T|) = z/2 \geq u$, since $\varepsilon|T| \geq 1$. If $G[T]$ is ε -sparse, the same bound gives a stable set of G . $\square$

9. THE FORMALIZATION

9.1. What is checked. The directory `formal/` of the repository <https://github.com/machin-e-qed/erdos-hajnal-six-vertex-path> contains a Lean 4 project that proves Theorem A. The file `formal/EHP6/Defs.lean` contains the target statement and the definitions it uses; they are quoted below. A graph is a `SimpleGraph` on a finite type, sizes are compared as real numbers, and G is H -free if no injective map from $V(H)$ to $V(G)$ preserves both adjacency and non-adjacency.

```
def pathGraph' (n : ℕ) : SimpleGraph (Fin n) := SimpleGraph.fromRel (fun i j => i.val + 1 = j.val)
def P6 : SimpleGraph (Fin 6) := pathGraph' 6

def ContainsInduced {n : ℕ} (H : SimpleGraph (Fin n)) (S : Finset V) : Prop :=
  ∃ f : Fin n → V, Function.Injective f ∧ (∀ i, f i ∈ S) ∧ ∀ i j, G.Adj (f i) (f j) ↔ H.Adj i j
def Free {n : ℕ} (H : SimpleGraph (Fin n)) : Prop := ¬ ContainsInduced G H Finset.univ

def IsCliqueF (T : Finset V) : Prop := ∀ a ∈ T, ∀ b ∈ T, a ≠ b → G.Adj a b
def IsStableF (T : Finset V) : Prop := ∀ a ∈ T, ∀ b ∈ T, a ≠ b → ¬ G.Adj a b

def EHforP6 : Prop :=
  ∃ τ : ℝ, 0 < τ ∧
    ∀ (V : Type) [Fintype V] [DecidableEq V] (G : SimpleGraph V) [DecidableRel G.Adj],
      Free G P6 → ∃ T : Finset V, (IsCliqueF G T ∨ IsStableF G T) ∧
        (Fintype.card V : ℝ) ^ τ ≤ T.card
```

The four imported theorems are the propositions `RodlCoP6`, `NssPath6`, `EhP5` and `NssComb` of `formal/EHP6/Imported.lean`, displayed next to their sources in Section 0.2. The main theorems, in `formal/EHP6/Final.lean`, are

```
theorem erdos_hajnal_P6_of_imported (hR : RodlCoP6) (hP : NssPath6) (hE : EhP5) (hC : NssComb)
  :
  EHforP6
theorem erdos_hajnal_P6_of_cited (hR : RodlCoP6) (hE : EhP5) (hC : NssComb) : EHforP6
```

The second follows from the first and `nss_path6_proof : NssPath6`, the formal proof of Theorem II in `NSSPath.lean`. The file `Audit.lean` prints the axioms on which each main lemma depends. For this theorem the output is

```
'EHP6.erdos_hajnal_P6_of_imported' depends on axioms: [propext, Classical.choice, Quot.sound]
```

and the same holds for `erdos_hajnal_P6_of_cited` and `nss_path6_proof`. These are the three standard axioms of Lean’s logic, so Lean checks outright that Theorems I, III and IV imply Theorem A. The file `Axioms.lean` asserts these three propositions as axioms, citing the literature, and `EHP6.erdos_hajnal_P6 : EHforP6` is the main theorem applied to them; its axiom list consists of the three standard axioms and these three. To trust the formal result one has to trust the Lean kernel, the definitions above, and the propositions `RodlCoP6`, `EhP5` and `NssComb` of `Imported.lean`. No proof in the project uses `sorry`.

The file `DefsCheck.lean` ties these definitions to Mathlib’s standard ones. It proves that `P6` is Mathlib’s `pathGraph 6`, that `Free G H` holds exactly when there is no induced embedding of H into G in Mathlib’s sense (`SimpleGraph.Embedding`), that `IsCliqueF` and `IsStableF` agree with Mathlib’s `IsClique` and `IsIndepSet`, and that `Sparse G x S` says that the induced subgraph on S has maximum degree at most $x|S|$. It then restates the main theorem using Mathlib’s definitions only, again with no axioms beyond the three standard ones.

9.2. Correspondence. Table 2 lists, for every numbered statement of this paper, the Lean theorem that proves it. Lean statements are sometimes slightly more general, for example stated for a vertex set S of a larger graph rather than for a graph, or with lengths and widths as real numbers.

Table 2: Paper to Lean.

paper	Lean theorem	file
Imported Theorems I–IV	<code>RodlCoP6</code> , <code>NssPath6</code> , <code>EhP5</code> , <code>NssComb</code> ; I, III, IV asserted as <code>rod1_coP6</code> , <code>eh_P5</code> , <code>nss_comb</code>	<code>Imported.lean</code> , <code>Axioms.lean</code>
Imported Theorem II, proved	<code>nss_path</code> , <code>nss_path6_proof</code>	<code>NSSPath.lean</code>
Lemma 0.3	<code>nss_L41_proof</code>	<code>NSSL41.lean</code>
Lemma 0.4	<code>nss_L42_proof</code>	<code>NSSL42.lean</code>
Lemma 0.5	<code>exists_anticonn_subset</code>	<code>Round2Util.lean</code>
modular decomposition, Lemmas 0.6, 0.7	<code>strong_laminar</code> , <code>module_pure</code> , <code>children_anticomplete_of_not_conn</code> , <code>children_complete_of_not_conn_compl</code> , <code>module_in_child_of_prime</code>	<code>MD.lean</code> , <code>MDGallai.lean</code>
the six-vertex certificate (Figure 1)	<code>coP6_of_facts</code>	<code>Certificate.lean</code>
Lemma 1.1 (module law)	<code>module_law</code>	<code>Local.lean</code>
Lemmas 1.2, 1.3	<code>diagonal_lemma</code> , <code>house_lemma</code>	<code>House.lean</code>
Corollary 1.4	<code>light_pattern_house_free</code> , <code>light_hom</code>	<code>House.lean</code> , <code>Round2Hom.lean</code>
Lemma 2.1 (Tooth Lemma)	<code>tooth_lemma</code> ; Steps 1–4: <code>antichain_mass</code> , <code>heavy_chain</code> , <code>node_step</code> , <code>chain_blockade</code> ; (F1): <code>sum_atomPart_eq</code>	<code>Tooth.lean</code> , <code>Tooth*.lean</code>
Lemma 2.2 (Pair Lemma)	<code>pair_lemma</code>	<code>Pair.lean</code>
Lemma 3.1	<code>comb_lemma</code> ; Step 1: <code>comb_extract</code> ; Steps 2–4: <code>comb_from_data</code>	<code>Comb.lean</code> , <code>CombExtract.lean</code>
Lemma 4.1	<code>round1_step</code>	<code>Round1.lean</code>
Lemma 4.2	<code>round1</code>	<code>Round1.lean</code>
Lemma 4.3	<code>round1_blockade</code>	<code>Round1.lean</code>
Theorem 5.1	<code>layout_theorem</code> ; (L3) under refinement: <code>ref_L3</code>	<code>Layout.lean</code>
Lemma 5.2	<code>nice_P6</code>	<code>Nice.lean</code>
Lemma 6.1	<code>round2_step</code> ; Steps 2, 3: <code>pass1_full</code> , <code>pass2_full</code> ; the claim in Step 4: <code>light_small</code>	<code>Round2.lean</code> , <code>Round2Pass.lean</code> , <code>Round2Hom.lean</code>
Lemma 6.2	<code>round2</code>	<code>Round2b.lean</code>
Definition 7.1, Corollary 7.2	<code>CruxC</code> , <code>crux_P6</code>	<code>Defs.lean</code> , <code>Crux.lean</code>

paper	Lean theorem	file
Lemma 8.1	c01_lemma1	C01Lemma1.lean
Lemma 8.2	c01_lemma2; the induction: c01_tree	C01Tree.lean
Theorem 8.3	polyRod1	Final.lean
Theorem A	eh_of_polyRod1, erdos_hajnal_P6_of_imported, erdos_hajnal_P6_of_cited, erdos_hajnal_P6	C01EH.lean, Final.lean

9.3. Independent checks. Lean’s kernel is small, and tactics can only produce proof terms that the kernel re-checks, so an error in a tactic cannot make a false statement pass. Kernel bugs are the remaining risk. Lean versions up to 4.33.0 have known soundness bugs, all requiring deliberately constructed input. The project was first written for Lean 4.15.0 and was then ported, changing proof scripts only, to Lean 4.34.1, the current release, in which all of them are fixed. The following checks were run on that version.

- (1) *Build and audit.* Every file is elaborated and checked by the kernel, and the axiom audit gives the output above.
- (2) *Fresh replay.* `leanchecker --fresh EHP6.Final` re-checks, starting from an empty environment, every declaration on which the final file depends, Mathlib included. This takes about eleven minutes.
- (3) *Comparator.* The comparator tool [13] builds a trusted file, `Challenge.lean`, that states both main theorems with no proof and imports only `Defs.lean` and `Imported.lean`. It checks that the project proves exactly these statements using only the three standard axioms, and replays the exported proof in the Lean kernel.
- (4) *Independent kernel.* Comparator also passes the exported proof to nanoda [14], a type checker written in Rust that shares no code with Lean’s kernel. It accepts the proof.

As negative controls, comparator rejects the same statement with the hypothesis `hE` removed (“statement do not match”) and rejects `erdos_hajnal_P6` when only the standard axioms are permitted (“Illegal axiom detected”). The project contains no `unsafe` code, no `native_decide` and no metaprogramming. For a false proof to pass checks 3 and 4 it would have to fool two unrelated kernels at once.

9.4. Reproducing the checks. The project uses Lean v4.34.1 (file `formal/lean-toolchain`) and Mathlib at tag v4.34.1, pinned in `formal/lake-manifest.json` to the commit

d13f23b723b8a846827a245b89c10fc7d3f11612.

With `elan` installed, run in `formal/`:

```
lake exe cache get      # download the compiled Mathlib
lake build              # check every file of the project
lake env lean Audit.lean
lake env leanchecker --fresh EHP6.Final
```

The repository runs the build, the audit, the fresh replay, comparator and nanoda on every push in GitHub Actions (`.github/workflows/lean.yml`), with every tool pinned to a fixed commit. The job fails if any `sorry` occurs, if an axiom other than the three standard ones appears in `erdos_hajnal_P6_of_imported`, `erdos_hajnal_P6_of_cited` or `nss_path6_proof`, or if either kernel rejects the proof. The file `VERIFICATION.md` records the tool versions and outputs, and the release notes record the commit and checksums of the released version.

9.5. Two corrections made during formalization. The formalization found two errors in earlier versions of the argument, both in the statements of cited results rather than in the new part of the proof. First, an early version assumed Lemma 0.4 as an axiom without the hypothesis

$A \neq \emptyset$; in that form the statement is false (take $A = \emptyset$ and $B \neq \emptyset$). The hypothesis was added, and both Lemma 0.3 and Lemma 0.4 are now proved in the project rather than assumed. Second, the source of the comb lemma was misattributed in an early draft; Theorem IV is [10, Lemma 4.3], a special case of a lemma in [4]. Each of the four imported propositions was checked against the L^AT_EX source of the cited paper, and Theorem II was then proved in the formalization.

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The architecture of the proof is that of the proof for P_5 by Tung Nguyen, Alex Scott and Paul Seymour [10]. The proof relies on theorems of Vojtěch Rödl [11], of Nguyen, Scott and Seymour [9, 10], and of Maria Chudnovsky, Alex Scott, Paul Seymour and Sophie Spirkl [4]. The formalization uses Lean [19] and Mathlib [18], and was checked with `leanchecker` (Kim Morrison and Sebastian Ullrich), `comparator` and `lean4export` (Lean FRO) [13], and `nanoda` [14]. Review comments from ChatGPT (OpenAI), passed on by the owner of the repository, led to several corrections, among them the move to a current Lean release and the check with an independent kernel. None of the people named here has reviewed this paper.

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